# Talk abstracts

## Session 1 — Congenital and Developmental Disorders

### *Thyroid hormone regulates axis patterning, epidermis and orientation of zebrafish fins*

Sarah McMenamin

Boston College

Thyroid hormone is a global endocrine factor that orchestrates developmental coordination across vertebrate species; in a medical context, thyroid disorders cause both metabolic and developmental pathologies. We have demonstrated that thyroid hormone regulates the skeletal patterning of fin rays along the proximodistal axis and establishes the adult position of the paired fins in zebrafish. Thyroid hormone is both necessary and sufficient to stimulate distal features in the fin rays, and this role is independent and may be decoupled from the processes that regulate ray growth and ultimate fin ray length. Furthermore, we show that thyroid hormone regulates the growth of the epidermal layer of soft tissue surrounding the fins. Limbs rotate during early human embryogenesis. Similarly, zebrafish pectoral fins—which are evolutionarily homologous to tetrapod forelimbs—rotate relative to the body axis during larval development to become horizontally oriented in the adult. We show that thyroid hormone titer gates both the growth of the pectoral fin rays and the rotation of the fin relative to the underlying girdle. Thyroid hormone coordinates multiple morphogenetic roles in shaping fins, and our findings demonstrate how this conserved endocrine pathway can be modulated to drive the diversification of functional characteristics of appendages.

### *The characterization of neural crest migration defects in Nager/Rodriguez syndromes*

Casey Griffin, Mauricio De La Rosa, Francesca Torres, Jean-Pierre Saint-Jeannet

New York University

Nager and Rodriguez syndromes are rare craniofacial and limb disorders characterized by midface retrusion, micrognathia, absent thumbs, and radial hypoplasia. These disorders result from haploinsufficiency of SF3B4 (splicing

factor 3b, subunit 4), a component of the pre-mRNA spliceosomal machinery. While the spliceosome is present and functions in all cells of the body, most spliceosomopathies – including Nager/Rodriguez syndromes – are cell/tissue-specific in their pathology. Previous work developed a Xenopus tropicalis sf3b4 mutant line using CRISPR/Cas9 gene editing. We showed that homozygous deletion of sf3b4 is detrimental to the development of cranial neural crest-derived cartilage progenitors. Temporal RNA-sequencing analyses of mutant embryos identified transcriptional changes associated with an enrichment of terms consistent with defects in neural crest cell migration and survival. Recently, we have identified these key changes and are beginning validation and characterization of these genes to unravel the molecular mechanisms underlying the pathogenesis of Nager/Rodriguez syndrome, including genes associated with migration (NRP1, Tbx1) and extracellular matrix organization (Vitrin, MMP7, Adamts1). With this characterization, we can begin to understand the pathomechanisms underlying spliceosomopathies, which can translate to better point of care for patients in the long term.

## *DYRK1A regulates nephron progenitor survival and kidney development in Xenopus laevis*

Adrian Romero, Mark E. Corkins, Rachel K. Miller

UTHealth Houston

Congenital anomalies of the kidney and urinary tract are a leading developmental cause of pediatric renal failure, yet the mechanisms controlling nephron development remain incompletely understood. Haploinsufficiency of the dosage-sensitive kinase DYRK1A is associated with structural kidney abnormalities, but its role during nephrogenesis is unclear. We investigated how Dyrk1a regulates nephron progenitor survival and tubule morphogenesis in Xenopus. Dyrk1a showed dynamic subcellular localization, with strong nuclear enrichment in early Lhx1-positive nephron progenitors and increased cytoplasmic and membrane localization during epithelial differentiation. Dyrk1a knockdown reduced kidney size and progenitor number. Morpholino-mediated depletion and pharmacological inhibition increased apoptosis without altering proliferation. CRISPR/Cas9-mediated disruption of dyrk1a reproduced the increased apoptosis and kidney developmental defects. A phosphotyrosine-binding domain microarray identified activation-dependent interactions between autophosphorylated DYRK1A and PI3K signaling components, including the p85 regulatory subunit. Consistent with this finding, DYRK1A inhibition reduced AKT phosphorylation, disrupted apical

pAKT localization, and progressively impaired proximal tubule formation. High-dose inhibition prevented tubulogenesis and retained Pax8-positive cells in an undifferentiated state. DYRK1A inhibition also reduced RNA polymerase II CTD phosphorylation, while single-cell RNA sequencing revealed depletion of a progenitor-derived population committed to epithelial differentiation. Together, these findings identify DYRK1A as a dosage-sensitive regulator of nephron progenitor survival and tubulogenesis and support a functional link between DYRK1A and PI3K–AKT signaling.

## *Zebrafish models of imperforate anus and anorectal malformations*

Brent Hawkins, Lisa Schimmenti, Matthew Harris

Boston Children's Hospital

The digestive system of humans normally terminates in an anal opening that is independent from the urinary and reproductive exits. This separated configuration arises from a transient period during embryogenesis in which digestive and urogenital tracts share a common lumen called the cloaca, and over the course of development these tracts are resolved into their respective channels. However, in ~1 in 3,000 pregnancies the partitioning of these tracts is incomplete resulting in anorectal malformations (ARMs). ARMs exhibit a spectrum of severity ranging from imperforate anus to full cloacal malformation and likely have complex and varied genetic etiology. To interrogate the genetic and developmental basis of ARMs we are pioneering the zebrafish as a powerful in vivo model. Larval zebrafish exhibit a “pseudo-cloacal” arrangement in which the digestive and urinary systems exit the body in adjacent but separate openings. These systems are homologous to their counterparts in mammals and share genetic regulatory logic. The external development and optical clarity of the zebrafish permit the real-time tracking of tract formation and rapid phenotyping. Through genetic perturbation we have induced aberrant tract fusion (cloacal malformation) as well as distraction of the openings to model a fully independent anus in the zebrafish. New precision genome editing tools allow us to rapidly model in zebrafish candidate mutations identified in patients to investigate their consequence on tract patterning. We are currently analyzing genetic data from patient-parent trio to identify candidate mutations to model using the zebrafish platform.

## *Unraveling the genetic, cellular, and tissue-level mechanisms of vertebrate enteric nervous system construction*

Rosa A. Uribe

Rice University

The enteric nervous system (ENS), often called the “second brain,” is the largest branch of the peripheral nervous system and plays a vital role in gut health. This vast network of nerves is embedded in the muscle walls of the digestive tract and controls many critical functions, such as peristalsis (gut motility), fluid balance, and hormone secretion. It also communicates directly with the brain through the spinal cord, making it a key player in the gut-brain axis. Despite its importance, we still don’t fully understand how the ENS forms during development or how its dysfunction contributes to disease. Disruptions in ENS development are at the root of Hirschsprung disease (HSCR), a condition in infants where parts of the colon lack enteric neurons, requiring surgery and often resulting in ongoing complications such as Hirschsprung-associated enterocolitis (HAEC).

My lab leverages zebrafish as a powerful vertebrate model to tackle key questions in ENS developmental biology. Our team pioneered the first ENS single-cell RNA-sequencing transcriptomic atlas in zebrafish, 4D imaging of whole-gut ENS assembly in vivo, whole-gut Spatial Genomic Analysis (SGA), which is a 3D multiplexed in situ assay spatially mapping ENS molecular regionalization, and whole animal CRISPR-Cas9 mutational screening that uncovered novel ENS regulators. For this talk, I will highlight how we are building on these recent advances, with the aim of explaining how the ENS is built and eventually pioneer new strategies to repair or restore it in disease settings.

## Keynote Lecture 1

### *A regeneration-specific blastema integrates developmental and regenerative gene regulatory programs in sea star larvae*

Jon Lee Andrade, Andrew Wolff, Veronica Hinman

University of Florida

Whether regeneration depends on the reactivation of developmental programs, regeneration-specific regulatory mechanisms, or both remains a central question in regeneration biology. We address this question using regenerating larvae of the sea star Patiria miniata, a deuterostome with robust whole-body regenerative capacity. By integrating single-nucleus transcriptomics with chromatin accessibility profiling across development and regeneration, we identify a regeneration-induced blastema cell state that is molecularly distinct from pre-existing larval populations and serves as the source of regenerated tissues. We further identify distinct classes of regeneration-responsive enhancers that connect wound-induced signals to developmental gene regulatory networks, including enhancers shared with embryogenesis and others deployed specifically during regeneration. Together, these findings suggest that regeneration in Patiria miniata is orchestrated through a specialized regenerative program that integrates developmental and regeneration-specific gene regulatory networks, providing a new framework for investigating the mechanisms and evolution of whole-body regeneration.

## Session 2 — Regeneration and Reproductive Biology

### *Dynamic chromatin loop rewiring underlies sensory epithelial regeneration within a stable higher-order genome architecture*

Luis Colón-Cruz, Guanrong Yan, Aranza Torrado, Rebecca Landry, Ava Edison, Shawn Burgess

National Human Genome Research Institute

Sensory hair cell (HC) loss in the mammalian inner ear leads to permanent hearing impairment, whereas non-mammalian vertebrates such as zebrafish retain the capacity to regenerate these cells through injury-induced reprogramming. We used nucleosome-resolution Micro-C to generate high-resolution chromatin and time-resolve contact maps of the adult zebrafish inner ear under homeostatic conditions and across regeneration at 4-, 5-, and 7-days following diphtheria toxin-induced HC ablation. We found that during regeneration, large-scale genome architecture remained broadly stable. A/B compartment remodeling was limited, chromosome-specific, and temporally phased, while global TAD organization remained constrained. In contrast, chromatin loops represented the most dynamic architectural layer. Loop analyses revealed a shared structural scaffold across conditions together with stage-specific interactions, with the most prominent unique loop class emerging at late regeneration. Integration of these 3D genome maps with transcriptional data from mature ears strengthened the analysis by showing that loop gain, and especially loop turnover, were associated with increasingly distinct gene expression behavior, while loop-remodeled genes were enriched for transcription factors and developmental and sensory morphogenesis programs. Targeted perturbation of distal regulatory elements and CTCF-associated architectural sites across the sox2 domain impaired hair cell regeneration in graded and position-dependent ways, demonstrating that chromatin topology is not merely correlative but mechanistically required for tissue repair.

### *Scaling the regenerating axolotl limb: regulation of allometric skeletal growth*

Shauna Kelly, Emily Wang, and Catherine McCusker

University of Massachusetts Boston

Axolotls regenerate limbs that are precisely scaled to the size of the body, restoring both form and function following injury. While the cellular

mechanisms that regulate the developmental stages of regeneration are increasingly understood, far less is known about how the limb regenerate achieves its final size. Following blastema formation and patterning, regenerating axolotl limbs initially emerge undersized and subsequently undergo rapid allometric growth until their proportions are restored. We previously demonstrated that limb nerves are required for this growth phase and that their growth-promoting activity depends on connectivity to the central nervous system (CNS), suggesting the existence of an upstream systemic regulator. Here, combining large-scale transcriptional profiling with in vivo functional analyses, we identify neuroendocrine signaling as a CNS-derived regulator of regenerative limb growth. Our findings further reveal a signaling hierarchy in which a neuroendocrine cue acts through ERK signaling in limb nerves to activate BMP signaling within the regenerate, thereby promoting long bone growth during regeneration. Together, these results uncover a neuroendocrine–neural–skeletal signaling axis that links the CNS to the control of regenerative growth and provide new mechanistic insight into how axolotls restore limbs of the correct size following amputation.

## *Early embryonic shape directly regulates development*

Chuan Qin, Hui Chen

University of South Carolina

Irregularly shaped human oocytes are associated with poor embryonic development in assisted reproduction; however, whether and how early embryo shape directly regulates development remains unclear. Here, we establish Xenopus early embryos as a system to investigate shape-regulated development. Compared with controls, naturally occurring irregular embryos exhibit striking early defects, from embryonic lysis to impaired tissue formation, consistent with clinical observations in human embryos. To test whether shape directly affects development, we altered embryo shape by removing the vitelline membrane at the 1-cell stage, causing spontaneous flattening. Shape-altered (SA) embryos displayed early defects resembling naturally irregular embryos. Altering shape at different stages revealed stage-dependent effects, with earlier perturbations causing more severe defects. Although SA embryos self-organized into spheroids before the mid-blastula stage, this recovery was insufficient for normal development, demonstrating an essential role for early embryonic shape. Confocal imaging revealed large cytoplasmic regions with few nuclei in the vegetal hemisphere of SA embryos, suggesting abnormal cell division patterns. Transcriptome analysis identified hundreds of differentially expressed genes during zygotic genome activation

involved in cell differentiation and tissue morphogenesis. To test whether restoring shape rescues development, we designed hemispherical agarose wells to constrain SA embryos. Only wells closely matching normal embryo size restored spherical shape and normal development. Mechanistically, Wnt signaling was highly activated in SA embryos, and the Wnt inhibitor XAV-939 rescued normal development, implicating Wnt signaling in shape-mediated development. Together, these findings demonstrate that embryonic shape directly regulates development and suggest restoring early embryonic shape may improve human assisted reproduction.

## *The wild zebrafish sex determination mechanism inhibits the feminizing effects of mutations in androgen receptor and dmrt1 genes*

Angel Amores, Catherine Wilson, Yilin Yan, John Bui, Kimberly Ake, John Dowd, and John Postlethwait

University of Oregon

Men today are half the man their grandfather was. Globally, unselected men have half the sperm count as 50 years ago, although Americans may have bucked this trend. In young men, testicular germ cell tumors (TGCT) recently increased ~72%. These problems likely originate during fetal development but become evident decades later. In women, polycystic ovarian syndrome (PCOS) is the major cause of infertility, and like TGCT, initiates in prenatal development, likely related to anti-Müllerian hormone or testosterone. Environmental exposures can exacerbate these problems. Developmental genetic and toxicogenetic analyses in aquatic models can reveal mechanisms leading to diminished human reproductive health. Wild zebrafish have a ZZ male / ZW female chromosomal sex-determining system: ZZ gonads always develop directly into testes, and ZW gonads always initiate development as ovaries, but some secondarily become testes. To understand how chromosomal sex works in zebrafish, we asked: Is a ZZ genotype epistatic to feminizing mutations? Or vice versa? We induced mutations known to feminize domesticated zebrafish in the androgen receptor (ar) and the testis factor dmrt1. Results showed that, while nonsense mutations increased the number of ZW fish that retain ovaries, they did not cause ZZ gonads to make ovaries. We conclude that the ZZ genotype causes gonads to initiate a firm testis pathway not derailed by these otherwise feminizing mutations. Further studies into the molecular genetic basis of the ZZ / ZW sex-determination system, including studies with endocrine-disrupting chemicals, will illuminate

our understanding of how developing gonads initiate later problems, including TGCT and PCOS.

## *Epithelial organization in proliferating tissues: insights from sea anemone embryos*

Katerina Ragkousi

Department of Biology, Amherst College, Amherst, MA 01002, USA

The epithelial layer is a selective barrier and architectural scaffold encountered in tissues across all Metazoa. Cell assembly into epithelia occurs early during animal development and is maintained throughout proliferative growth despite dramatic changes in cell shape and cell contacts during cell division. We use embryos of the early branching sea anemone (Nematostella vectensis) to examine how a highly proliferative tissue organizes and maintains its epithelial architecture during development. In addition, Nematostella can organize into animals after tissue dissociation and reaggregation allowing us to examine how epithelia emerge from disintegrated embryos. We find that Nematostella embryos organize into epithelial monolayers and undergo periodic compactions early during development. In these embryos, however, cell divisions interfere with the integrity of the epithelial barrier. Apical polarity complexes appear to disengage from the cell cortex during cell division, and the embryonic epithelial barrier is disrupted. On the other hand, we find lateral polarization to be essential for epithelial organization in both intact embryos and cell aggregates from dissociated embryos. We currently examine how highly conserved polarity proteins aid cells to couple their intracellular organization to robust tissue-level order. These studies also provide insights into the origins and mechanisms of metazoan epithelialization.

## Session 3 — Cancer and Aging

### *Transcriptional and mitochondrial signatures of intergenerational and transgenerational inheritance of maternal age*

Nelia Luviano, Alyssa Liguori, Sovannarith Korm, and Kristin E. Gribble

Marine Biological Laboratory

Maternal age effects, in which the age of a mother affects offspring phenotype without a change in genotype, have been observed across species for a century. The molecular controls of this “non-Mendelian” inheritance of age are unknown, however. To reveal the mechanisms of maternal age effects, we reared lines of Brachionus manjavacas rotifers of constant young or old maternal age for three generations. In the third generation, we assessed lifespan, reproduction, and gene expression (RNA-Seq) in offspring from young mothers and grandmothers, old mothers and grandmothers, and in “switched” maternal-age cohorts. Lifespan and fecundity were lower in old-mother lines than in young-mother lines. Switching maternal age resulted in intermediate lifespan, demonstrating transgenerational inheritance of maternal age effects. Upregulation of oocyte maturation and cell cycle regulation genes in 1-d-old offspring in old-mother lineages suggests accelerated reproduction is associated with shorter lifespan. Maternal age- and age-associated shifts in expression of histone modifiers suggests increased transcription in old-mother lines, which may lead to loss of coordination of biochemical pathways. Findings from RNA-Seq, in conjunction with prior mitochondrial metabolomics results, suggest that metabolic pathways, including methionine/cysteine, purine, and pyrimidine metabolisms, are upregulated at young ages but downregulated more rapidly in late life in old-mother lines compared with young-mother lines. Together, these results suggest that old-mother offspring have accelerated development and early life reproduction at the expense of longevity. These effects may be regulated through changes in mitochondrial metabolism which alters metabolites available for histone modification, causing changes in gene expression.

## *A functional genetic screen of neuropeptides for vertebrate longevity*

Ravi D. Nath, Claire N. Bedbrook, Daniel Heinzer, Felix Boos, Rogelio Barajas, Karl Deisseroth, Anne Brunet

Princeton University

Aging causes systemic decline and leads to an increased probability of disease. Identifying new strategies to promote healthy aging is essential to counter age-related dysfunction and disease. Small peptides are exciting candidates for the systemic regulation of aging. However, the long timescale and complexity of vertebrate aging have made the discovery of new longevity peptides challenging. Here we used a naturally short-lived vertebrate, the African killifish, to conduct the largest screen of vertebrate longevity to date. We focused on 22 neuropeptide-encoding genes that are conserved in humans, because neuropeptides regulate key physiological processes—such as sleep, metabolism, and stress—that influence long-term health. Notably, we discovered six neuropeptide genes that, when mutated, regulated lifespan. We focused on Neuromedin U mutants because mutation of this gene resulted in the largest lifespan extension in our screen. We discovered that Neuromedin U mutants were behaviorally youthful and exhibited changes to growth, metabolism, and inflammation pathways, all of which may coordinate impact on lifespan. These results highlight the power of the killifish for screening vertebrate lifespan and reveal that neuropeptides may represent an important class of vertebrate longevity factors.

## *Sex-, morph-, and age-associated transcriptomic differences in Xiphophorus multilineatus skeletal muscle*

Abby Quimby, Dr. Peggy Biga

Texas State University

Sex and age are strong determining factors of muscle growth, metabolism, and tissue maintenance yet the molecular mechanisms underlying these sex-biased differences remain poorly understood. Xiphophorus multilineatus provides a unique model for examining the genetic sex-effect versus physiology in skeletal muscle aging because males occur as distinct reproductive morphs, either sneaker-strategy or courter-strategy. These polymorphs cause differences in morphology, behavior, and skeletal muscle growth – as sneaker males are much smaller than courter males. Here, we used RNA sequencing to characterize sex-, morph-, and age-associated

differences in skeletal muscle gene expression in young and old female, sneaker male, and courter male X. multilineatus. Transcriptomic patterns were examined using principal component analysis and differential gene expression followed by Gene Ontology enrichment analysis.

- Sneaker males exhibited a stronger age-associated transcriptomic pattern, with samples clustering according to age, whereas courter males and females showed less distinct age-related clustering.
- Courter males showed diverse transcriptomic patterns and appeared to cluster closer with females of similar age rather than sneaker males. Aging in sneaker males was associated with changes in pathways related to cytoplasmic transcription and protein formation.
- These preliminary results suggest that skeletal muscle aging is not transcriptionally uniform across sexes and male morphs in X. multilineatus.

Further this suggests that sex on its own, is not responsible for age-associated transcriptomic changes in skeletal muscle, rather reproductive strategy has a stronger effect on aging. Whether this is due to differences in genetic control, behavior, or epigenetic regulation has yet to be determined.

## *Ancient defenses: genotoxic stress activates the DNA damage response and innate immune pathways in sea urchin coelomocytes*

Riss M. Kell, Margaret K. Weber, Rosa Y. Escalante, Jeffrey C. Silva, Andrea G. Bodnar

Gloucester Marine Genomics Institute

Mechanistic crosstalk between the DNA-damage response (DDR) and the innate immune system is essential to maintain genomic integrity, tissue homeostasis, and organismal resilience, yet its evolutionary origins remain poorly understood. As a long-lived, cancer-resistant basal deuterostome, the purple sea urchin Strongylocentrotus purpuratus provides a powerful system to investigate the ancestral origins and complexity of DDR-immune system crosstalk. Furthermore, while coelomocytes (the innate immune cells of sea urchins) have been shown to possess a low susceptibility to DNA damage in vitro, it is unknown if specific coelomocyte populations differ in their susceptibility to DNA damage. Here, we used S. purpuratus to identify immune-cell-intrinsic responses to genotoxic stress. UVB-induced DNA damage elicited a robust transcriptional response in S. purpuratus coelomocytes, activating conserved DDR genes in addition to autophagy,

ubiquitin signaling, and innate immune pathways. Single-cell RNA sequencing performed six hours post UVB exposure identified phagocytes and vibratile cells as the major mediators of the response to UVB challenge. Functional assays confirmed increased autophagy, while western blots and PTM-proteomics revealed widespread post-translational modification (ubiquitination and phosphorylation) of substrates in key signaling pathways. Our findings demonstrate the concurrent induction of DDR and innate immune pathways in an invertebrate deuterostome and establish a molecular framework for understanding how echinoderms respond to damaged self in the absence of vertebrate-specific DNA sensors. By revealing the conserved mechanisms that link the DDR with innate immunity, this work lays the foundation for comparative studies of damage recognition systems that shape deuterostome defenses, promoting genomic stability and cancer resistance.

### *From pools to pathways: Ecology becomes biologically embedded to shape vertebrate aging*

Enoch Ng'oma, Tamanna R Joya, Andrew Thompson, Peyton K Warren, Priyanka Patel, Ellen Mazalale

University of Missouri–Columbia

Aging is typically investigated under highly controlled laboratory conditions, yet aging evolved in complex and fluctuating natural environments. Understanding how ecological history shapes aging therefore requires integrating evolutionary, ecological, and molecular perspectives rather than studying isolated biological mechanisms. Seasonal African killifishes (Nothobranchius) provide a unique vertebrate system for addressing this challenge because they combine naturally evolved variation in lifespan with exceptional experimental tractability and spontaneous development of numerous age-associated phenotypes. In this talk, I will present an ecological systems framework for understanding vertebrate aging that spans natural populations and experimental environments. I will first show how whole-genome sequencing, multivariate phenotyping, and gut microbiome analyses reveal hierarchical divergence across wild populations, with increasingly context-dependent biological organization from genomes to microbiomes. I will then describe ongoing work comparing laboratory and semi-natural mesocosm populations across the aging process using longitudinal transcriptomics, quantitative proteomics, microbiome profiling, and lifespan analyses to determine how environmental complexity becomes embedded across molecular and organismal levels. Collectively, these studies support the view that aging is an emergent property of interactions among biological

systems shaped by ecological experience throughout life. By linking evolutionary ecology with multi-omic analyses of aging, annual killifish provide a powerful vertebrate model for understanding natural variation in the aging phenotype while offering a natural platform for investigating age-associated diseases, including cancer.

## Session 4 — Aquatic Genomics and Genetic Approaches to Disease Discovery

### *Genetic mechanisms of cell fate canalization and resilience in the developing zebrafish*

Daniel E. Wagner

University of California, San Francisco

Embryogenesis generates diverse cell identities and highly stereotyped ("canalized") developmental outcomes, despite intrinsic and extrinsic challenges. My laboratory combines single-cell genomics and experimental embryology to investigate how cell identities arise and the mechanisms that preserve developmental programs against perturbation. Building on our longstanding efforts to map vertebrate development at single-cell resolution, we present ZMAP, a multi-study consensus atlas of zebrafish embryogenesis. Integrating datasets across laboratories, developmental stages, and single-cell technologies, ZMAP reveals a consensus hierarchy of cell identities and provides a framework for comparing developmental trajectories and cell-type-specific gene expression programs.

We and others have observed that zebrafish embryos can survive prolonged oxygen deprivation by entering a reversible state of suspended animation, followed by normal development upon recovery. How do embryos preserve developmental potential through such striking interruptions? Using ZMAP, single-cell profiling, and in vivo imaging, we show that hypoxia-induced pausing leads to the temporary collapse of Wnt, BMP, FGF, and Nodal signaling gradients, despite broad preservation of cell identity at the transcriptional level. Paused embryos also upregulate a "Dormancy-Enriched gene Expression Program" (DEEP) comprising genes not typically expressed during embryogenesis, including the non-canonical poly(A) polymerase, tent5ba. Genetic disruption of tent5ba is compatible with normal development but, following hypoxia, leads to reduced survival, impaired DEEP expression, and axial patterning defects upon developmental re-entry. Poly(A)-tail and transcriptional profiling further link tent5ba to the

maintenance of transcripts involved in metabolism, germ layer specification, and developmental patterning. These findings support a model in which post-transcriptional poly(A)-tail regulation safeguards developmental trajectories through suspended embryogenesis.

## *Zebrafish runx1-2A-creERT2 CRISPR knock-in lineage tracing of definitive hematopoietic stem cells from embryo to adult*

James A Preston, Masuma K Usha, Stephen C Ekker, Karl J Clark, Jeffrey J Essner, Raquel Espin-Palazon, Maura McGrail

Iowa State University

Zebrafish combines the power of genetics and unparalleled in vivo imaging for investigating the dynamics of vertebrate hematopoietic development. Across species, the transcription factor Runx1 is essential for definitive hematopoiesis. We generated a zebrafish runx1-2A-creERT2 CRISPR knock-in for tamoxifen-regulated Cre recombinase Runx1 lineage tracing and characterized its activity using the ubi:Switch recombinase-dependent fluorescence reporter, microscopic live imaging and flow cytometry. Tamoxifen treatment beginning at gastrula stage labeled all expected Runx1 lineages in the early embryo, including neuroectodermal olfactory placode and Rohan-Beard neurons, primitive hematopoietic blood cells, and nascent hematopoietic stem and progenitor cells (HSPCs) in the dorsal aorta. Runx1-positive HSPCs populated the larval caudal hematopoietic tissue beginning at 2 days post fertilization (dpf), with subsequent colonization of the larval thymus beginning at 3 dpf and the kidney at 5 dpf. Timed tamoxifen induction of Cre activity allowed separation of Runx1 primitive hematopoiesis from definitive HSPC emergence and larval stem cell niche colonization. Flow cytometry of kidney marrow and peripheral blood from adults treated with tamoxifen at gastrula stage revealed Runx1 embryonic hematopoietic cells contributed to adult hematopoietic precursors, myeloid, lymphoid, and peripheral blood lineages. Labeling of all blood lineages was also effective by tamoxifen treatment of 5-month-old adults. The zebrafish runx1-2A-creERT2 line provides a powerful genetic tool for precise spatial and temporal analysis of Runx1 progenitor mechanisms in developmental and adult hematopoiesis.

## *Protein language models resolve neuronal receptor repertoires across Mollusca*

Jorge L. Perez-Moreno, Paul S. Katz

University of Massachusetts Amherst

Protein language models are neural networks trained on millions of protein sequences, learning representations that capture protein structure and function directly from sequence. Trained this way, they recognize distant relationships that alignment-based methods miss, comparing proteins by learned features instead of raw sequence similarity. We apply them to molluscan neuronal receptors, a large group that remains poorly annotated. Two receptor classes that dominate human neuropharmacology, metabotropic G protein-coupled receptors and ionotropic ligand-gated ion channels, are also the hardest to annotate, because their rapid divergence breaks homology-based methods outside traditional model species. Fewer than twenty of the roughly 1,850 metazoan class-A GPCRs with curated function are molluscan, while over ten thousand remain unannotated. Our AI classifier pairs a contrastive projection head with model embeddings to recover receptor identities that conventional pipelines miss. It outperformed profile hidden Markov models and sequence-identity searches, and matched phylogenetic placement at a fraction of the computational cost. That efficiency matters most where it is hardest to gain: phylogenetic placement needs a reliable alignment and reference tree, and the rapid divergence typical of chemoreceptors degrades both. We resolved the chemosensory receptor repertoire of the nudibranch Berghia stephanieae, a stringent test given its divergence, validating against characterized receptors from molluscs, nematodes, and vertebrates. The same framework was extended to ionotropic glutamate receptors, nicotinic acetylcholine receptors, and related channels. Most neuroactive drug targets sit in these classes, many still orphaned, and annotating them supplies the foundation that deorphanization and comparative pharmacology require in any divergent aquatic model of disease.

## *Neurogenesis in the largest invertebrate nervous system*

Jessica Stock, Alyson Hally, Loyal Goff, Carrie Albertin

Harvard University

Soft bodied cephalopods (octopus, squid, and cuttlefish) are remarkable animals: they have shapeshifting, color changing skin, flexible, sucker-lined arms that can taste and manipulate objects, three hearts, blue blood, and complex brains. With approximately 500 million neurons, cephalopods

possess the largest nervous systems among invertebrates, rivalling those of vertebrates in both size and complexity. To investigate the developmental origins of their elaborate nervous systems, we combined lineage tracing, in situ hybridization chain reaction, live imaging, and CRISPR/Cas9-mediated genome editing in our model cephalopods, Doryteuthis pealeii and Euprymna berryi. We find that cephalopod nervous systems are derived from neurogenic niches located outside of the developing brain that require extensive long-range cell migrations to feed newly born neurons into the growing nervous system. This work uncovers a combination of spiralian features, vertebrate-like mechanisms, and evolutionary novelties that drive the development of the cephalopod brain and sheds light on fundamental principles for building a large nervous system.

## Keynote Lecture 2

### *Deciphering gene regulatory circuitry underlying cranial neural crest fate decisions*

Tatjana Sauka-Spengler

Stowers Institute for Medical Research

Abstract pending.

## Session 5 — Functional Genetic Screens in Disease

### *Genetic and pharmaco-behavioral screens identify suppressors of autism gene-associated phenotypes in zebrafish*

Priyanka Jamadagni, Yi Dai, Yunqing Liu, Hellen Weinschutz Mendes, April Pruitt, Suha Khan, Liang Yang, Tzu-Chieh Huang, Xiayuan Huang, P.J. Michael Deans, Novin Balafkan, Dejian Zhao, Gang Xu, Yihan Liu, Ningshan Li, Weimiao Wu, Sarah E. Fitzpatrick, Uma Neelakantan, Tianying Chen, Christina Szialta, David S. Jin, Cheryl M. Lacadie, Sheila Umlauf, Xenophon Papademetris, Yulia V. Surovtseva, Kristen J. Brennand, Zuoheng Wang, Ellen J. Hoffman

Yale School of Medicine

Advancing from >100 genes associated with autism spectrum disorder (ASD) to biological mechanisms and pharmacological candidates remains a central challenge. Here, we leverage the advantages of zebrafish as a highly scalable, in vivo vertebrate system to investigate the function of multiple ASD genes in parallel and identify novel pharmacological candidates. First, we established a pharmaco-behavioral database of 774 U.S. FDA-approved drugs in larval zebrafish. Using automated assays of visual-startle and sleep-wake behaviors, we identified unique behavioral fingerprints for 520 compounds spanning a range of clinical indications. Second, by mapping the behavioral fingerprints of zebrafish carrying mutations in nine large-effect ASD genes onto this database, we identified drugs that anti-correlate with each ASD gene mutant and mutant subgroup. Third, we performed targeted suppressor screens of anti-correlating compounds in zebrafish scn1lab and dyrk1a mutants, identifying estrogens, microtubules, mitochondria and lipid metabolism, as central pathways for further evaluation. Finally, we found that levocarnitine, a mitochondrial modulator and top suppressor of both mutant behavioral phenotypes, rescues regional brain activity deficits and dysregulated lipid metabolic pathways in mutants, as well as signaling deficits in human pluripotent stem cell-derived glutamatergic neurons carrying mutations in SCN2A and DYRK1A, providing evidence for conservation of drug rescue across systems. This study uncovers novel suppressors of ASD gene-associated phenotypes, highlighting mitochondrial and lipid metabolism as a potential target for individuals carrying mutations in specific ASD genes. Taken together, we show how zebrafish pharmaco-behavioral screens represent a powerful preclinical approach to illuminate novel drug candidates relevant to ASD genes.

## *Z-Terminator: systematic engineering of loss-of-function alleles in the zebrafish mitochondrially encoded proteome*

Ankit Sabharwal, Kyler S. Mitra, Md Roushan Ali, Jun Morisue, Jace Klein, Rishav Sarkar, Kaila Savage, Lucy Rai Thulung, Cassidy Petree, Amanda Zumbrock, Santiago Restrepo Castillo, Mireya Mota, Gaurav K. Varshney, Karl J. Clark, Eiko C. Ogiso, Stephen C. Ekker

Dell Medical School, The University of Texas at Austin

Pathogenic variants in the 13 protein-coding genes of the mitochondrial genome underlie clinically and biochemically heterogeneous disorders. Most mtDNA-encoded genes lack defined loss-of-function (LOF) models in vivo. Here we describe Z-Terminator, the first systematic in vivo atlas of loss-of-function alleles spanning all the mtDNA-encoded OXPHOS subunits in zebrafish (Danio rerio). We used constrained and unconstrained mitochondrial TALE base editors to introduce premature termination codon (PTC) alleles via C-to-T transitions across Complexes I, III, IV, and V. Editing was detected at every locus in vivo, with individual embryo heteroplasmy ranging from approximately 20% to greater than 70%. Larvae harboring PTC alleles displayed bioenergetic defects and elevated lactate. While mtDNA mutations are associated with sensorineural hearing loss, the cellular basis has remained unclear. Z-Terminator is the first vertebrate platform to directly link engineered mtDNA loss-of-function alleles to a heteroplasmy-dependent sensory phenotype. We show that engineered mtDNA LOF alleles directly impair hair cell function in proportion to heteroplasmy in a living vertebrate. Germline transmission and tissue-specific heteroplasmy distributions were confirmed for representative loci. Embryo-to-embryo heteroplasmy variation consistent with mitochondrial germline bottleneck dynamics was observed. Z-Terminator serves as a vertebrate platform for systematic, gene-specific study of mtDNA dysfunction in vivo and enables preclinical evaluation of therapeutic strategies for mitochondrial disease.

## *Topological definition of the Geometry of Life and Disease (GOLD) as a ground truth foundation for Digital Biology*

Keith C Cheng, Daniel J Vanselow, Alex Y Lin, Andrew Sugarman, David Northover, Weaver Goldman, Mee Siing Ngu, Khai C Ang, Joshua Warrick, Steven Y Wang, Stephen Senft, Roger Hanlon, Gregory Babbit, Earnest Fokoué, Patrick J La Riviere and Danny Chen

Penn State College of Medicine

Why computational whole-animal phenotyping? The cellular basis of life and disease was discovered through the histological study of normal and diseased tissues. Histology is the present gold standard for cancer and other tissue diagnostics, but is associated with volumetric undersampling, in addition to being unable to allow the computational measurement of 3D volumes, shapes and textures, or of the 3D relationships between all cells and structural elements. Advances in tomographic imaging allows us to imagine, for the first time, the possibility of practical, high-throughput phenotypic screens using 3D imaging. Complex phenotypes, caused by mutation and chemical exposures in model organisms, frequently affect more than one cell type and organ system at once. Parallel human and animal model tissue studies can validate models of human disease. Genetic and chemical screens (genetic and chemical phenomics) will require commitment to large-scale computational mechanisms for measuring cellular and tissue change in model organisms, across cell types and organ systems. Defining the Geometry of Life and Disease (GOLD) will allow us to advance our ability to understand development, physiology, and disease across phylogeny that will in turn allow us to modernize tissue diagnostics, understand gene function, create a safer drug, and enhance societal adaptability to new environmental and disease threats.

## *A conserved enhancer cluster drives dorsal retinal expression across vertebrates*

Lina Shi, Louise Perez, Gabriela Lima, Josane F. Sousa, Zhuoxin Chen, Evgeny Kvon, Igor Schneider, Patricia N. Schneider

Louisiana State University

The vertebrate retina is patterned along its dorsal-ventral (DV) axis by conserved signaling centers, yet the cis-regulatory architecture maintaining these territories remains unclear. Using the four-eyed fish Anableps anableps, which enables physical separation of dorsal and ventral retina, we integrated

RNA-seq, ATAC-seq and Hi-C to map DV regulatory landscapes. We identify a 29-kb dorsal-enriched enhancer cluster within a gene poor region between efnb2a and gtpbp8, embedded in a conserved topologically associating domain spanning efnb2a-gtpbp8-slc10a2. Dorsal-biased chromatin contacts link this cluster to the efnb2a promoter. The orthologous human region contains a genome-wide significant association with retinal thickness (rs2575134; P = 2e-14). Leveraging the compact genome of Tetraodon nigroviridis, we cloned the orthologous interval and show that it drives dorsal retina-specific reporter expression in zebrafish and mouse. Together, these results identify an evolutionarily conserved regulatory domain controlling retinal axial identity and linking enhancer architecture to human ocular variation.

## *Discovering the developmental basis of endosome trafficking disorders and congenital anomalies using Xenopus*

Nicole Edwards

University of Michigan

Endosome trafficking moves proteins between different cellular compartments. In adults, this pathway is important for cell physiology and membrane homeostasis, while in development it regulates cell shape changes, cell adhesion, and polarity. We have recently discovered that endosome trafficking is critical for organ morphogenesis using a Xenopus tropicalis model of human congenital trachea-esophageal malformations. We show that disruptions to endosome trafficking pathways disrupt epithelial cell-cell adhesion and cell polarity behaviors required for the embryonic foregut to septate into the trachea and esophagus. We will highlight the advantages of X. tropicalis as a disease model to investigate endosome trafficking-related congenital anomalies using recent gene editing and imaging technologies. Using F0 CRISPR-Cas9 gene editing, we can titrate loss of function indel mutation burden between heterozygote (no phenotype) and complete null (embryonic lethal) to functionally screen patient risk alleles for disease causality in X. tropicalis. We observe that gRNA design significantly influences gene editing outcomes rather than the dose of CRISPR reagents. We will also demonstrate different microscopy imaging modalities to phenotype the trachea and esophagus and visualize changes to subcellular processes caused by disrupted endosomal trafficking. Moving forward we will use X. tropicalis to investigate how epithelial cells use endosome trafficking pathways to drive tissue-level morphogenesis and determine the mechanisms underlying organ comorbidities associated with endosomeopathies.

## Session 6 — Metabolic and Endocrine Disease

### *Adaptations to rapid weight gain and loss in marine mammals*

Jane Khudyakov

University of the Pacific

Marine mammals have evolved an extraordinary capacity to store lipids in blubber, or modified subcutaneous adipose tissue, with body fat contents exceeding 40 percent in some large whales and polar seals. In fact, low fat stores are associated with poor health, reproduction, and survival outcomes in most marine mammal species. Migratory and fasting-adapted marine mammal species also show dramatic, rapid seasonal fluctuations in body fat. For example, adult elephant seals lose over 30 percent fat mass while fasting for one month during breeding, regaining most of their fat stores over four months of foraging at sea, during which energy gain may exceed 100 MJ per day. Our lab has been exploring the molecular mechanisms underlying healthy fat cycling and the response of blubber tissue to cortisol and thyroid hormones in two marine mammal "model" systems, the northern elephant seal and bottlenose dolphin. We have identified gene networks and proteins that characterize marine mammal blubber and how these factors respond to fasting, feeding, hormones, and contaminants. Our work provides potential insights into mechanisms that support metabolically healthy obesity in mammals, such as high adipogenic and antioxidant capacity and production of adipokines and lipoproteins involved in lipid turnover and insulin sensitivity.

### *Cell type-specific transcriptional responses to aryl hydrocarbon receptor (AHR) activation in larval zebrafish*

Christopher S. Murray and Neelakanteswar Aluru

Woods Hole Oceanographic Institution

The aryl hydrocarbon receptor (AHR) mediates cellular responses to environmental contaminants and plays important roles in neurodevelopment and immune regulation. Our previous work demonstrated that developmental exposure to the potent AHR agonist PCB126 alters microglial abundance and morphology in zebrafish, suggesting disruption of neuroimmune development. We investigated transcriptional responses to PCB126 in the zebrafish brain across life stages. Embryos were exposed to 2 nM PCB126 from 4–24 hours post-fertilization, and brains were analyzed by single-nuclei RNA sequencing (snRNA-seq) at 6 days post-fertilization. Analysis identified diverse neuronal cell populations and widespread induction of the canonical AHR target gene

cyp1a. Across neuronal cell types, PCB126 exposure was associated with increases in receptor-mediated and intercellular signaling functions and decreases in extracellular matrix, ribosome- and RNA-binding functions, pathways that may influence broader neuroimmune and neuron–glia interactions. In complementary studies, adult zebrafish were exposed to 10 nM PCB126 for 24 hours. Bulk RNA sequencing of whole brains identified 352 differentially expressed genes. Enriched pathways included xenobiotic response, neuroinflammation, blood-brain barrier function, and energy metabolism, with pronounced transcriptional responses associated with microglia. Microglial genes showed enrichment for immune-related pathways, indicating that acute AHR activation elicits a robust neuroinflammatory response. Together, these findings demonstrate that PCB126 activates AHR signaling across diverse brain cell populations while producing prominent effects on neuroimmune pathways. Integration of developmental snRNA-seq and adult transcriptomic data identifies AHR-mediated signaling as a potential mechanism linking environmental contaminant exposure to persistent and life stage-dependent alterations in brain immune function and neuronal homeostasis. [1R21ES034839-01; P01ES028938; OCE 2418297]

## *Nutrient-responsive gene regulatory programs underlying physiological adaptation in cavefish*

Tyler Guzman, Sumedha Mukhi, Vikas Tiwari, Edward Ricemeyer, Luliia Innokenteva, Rachel Carroll, Surendra Shukla, Narendra Singh, Wesley C. Warren, and Jaya Krishnan

Genes and Disease Research Program, Oklahoma Medical Research Foundation; Harold Hamm Diabetes Center and Department of Biochemistry and Molecular Biology, University of Oklahoma Health Sciences Center

Animals must continually adjust their physiology as nutrient availability changes. Understanding the gene regulatory basis of these dynamic responses requires organisms that have naturally adapted to nutrient extremes. The Mexican tetra, Astyanax mexicanus, provides one such powerful system because river-dwelling surface fish and independently evolved cave-dwelling cavefish experience contrasting nutritional environments and have evolved distinct physiology while retaining a shared vertebrate molecular toolkit.

To define the gene regulatory networks underlying adaptation to nutrient scarcity, we compared liver transcriptomes and chromatin accessibility profiles from fed and acute starved surface fish and cavefish. Integrated RNA-seq and ATAC-seq analyses revealed extensive remodeling of nutrient-responsive genes and cis-regulatory elements, with drastic differences in the

magnitude and direction of starvation responses between morphotypes. These data suggest that cavefish have evolved distinct regulatory programs associated with protein homeostasis, oxidative stress response, and mitochondrial function to cope with prolonged nutrient deprivation. Single-nuclear RNA sequencing reveals specific liver cell populations that mediate these responses to nutrient deprivation. Future studies in other metabolic tissues and nutritional states will distinguish tissue-specific from more broadly coordinated regulatory programs. Together, this provides a framework for understanding how gene regulation contributes to physiological plasticity and adaptation across nutrient extremes.

### *HCR imaging of diverse target types from tissue sections to whole-mount embryos*

Aneesh Acharya

Molecular Instruments

Imaging RNA and protein across diverse sample types requires high sensitivity, low background, and effective signal penetration that can hold up from thin tissues to intact embryos. This talk will introduce HCR™ imaging workflows designed to meet those challenges across sample formats.

We'll cover HCR™ HiFi Probes for RNA detection, HCR™ Gold for multiplexed RNA and protein imaging, and HCR™ Pro for highly sensitive RNA detection. For protein imaging, we'll highlight the HCR™ HiFi Encoder and new HiFi secondary antibodies, which pair with your trusted primaries to deliver stronger protein signal without requiring additional antibody optimization.

### *Nutritional reprogramming of DNA methylation signatures associated with fatty liver disease in Medaka*

Sourav Chakraborty, Santosh Anand, Ramji K. Bhandari

University of Missouri

Epigenetic inheritance of environmentally and lifestyle-induced phenotypes has been demonstrated in many model organisms in a laboratory setting, suggesting that current generations could be harboring ancestral environmental/diet exposure-induced molecular memories or future generations may carry such memories due to exposure of the current generation. Our laboratory has demonstrated that ancestral bisphenol A (BPA,

a ubiquitous chemical to which humans and ecological species are exposed worldwide) exposure can cause heritable NAFLD in medaka. This disease was found heritable across five generations with sex-biased pathology. Females showed severe NAFLD/ nonalcoholic steatohepatitis (NASH) phenotypes. Screening of DNA methylation profile of sperm revealed persistent inheritance of BPA-induced DNA methylation profile (epimutations) from ancestors to subsequent offspring generations. DNA epimutations predicted human NAFLD/NASH to happen in offspring. Transcriptional analysis of the offspring liver correlated with the predicted DNA epimutations. This inheritance started from F0-F1 generation and continued to F4 generations. Treatment of the F4 offspring with vitamin C (which acts as an antioxidant and epigenetically active compound) resulted in: 1) reversal of NAFLD phenotype to normal, 2) Correction of the majority of BPA-induced DNA epimutations, irrespective of their methylation status, and 3) restoration of transcriptional network comparable to that in untreated controls. NAFLD/NASH is a multifactorial disease with no FDA approved medications for treatment. The present results suggest that low-concentration nutritional epigenetic modifiers may be useful for correcting environmentally originated transgenerational liver disease opening avenues for further research towards understanding the role of epigenetic landscape in metabolic health and nutritional mitigation of NAFLD in humans.

## LSU AgCenter NAM Community Development

### *Interdisciplinary technologies for accessible and reproducible new approach methodologies*

Jack C. Koch, Phi Hoang, Yue Liu, Irene Marti-Gil, Nicholas Coxe, Marney A. Preston, Carl Anderson, Victoria K. Gibbs, Jeri Brandom, Stephen Watts, Zoltan M. Varga, and Terrence R. Tiersch

LSU Agricultural Center

WARNING: Do not attend this workshop if you are not seeking hands-on, interactive, and potentially career-changing experiences with emerging, open-source, interdisciplinary technologies supporting New Approach Methodologies (NAMs)! An extraordinary opportunity is emerging at the intersection of biomedicine, engineering, open-source technology, quality management, and genetic resources. Integrating these areas will help make NAMs more accessible, reproducible, and transferable between laboratories, while strengthening germplasm repositories that safeguard valuable genetic resources developed through decades of research, hundreds of careers, and with billions of government funding. The Aquatic Germplasm and Genetic Resources Centers (AGGRC) are world leaders in development and application of open-source hardware. With more than 15 years of experience in additive manufacturing, AGGRC has developed extensive capabilities using 3-D printing and has pioneered, among others, the transition from photolithography-based microfluidic devices toward accessible, customizable, resin-printed microfluidic platforms. These advances provide powerful opportunities to rapidly design, prototype, modify, and distribute user-friendly platforms to support diverse biomedical models and applications. Such platforms can catalyze innovation, advance rigor and reproducibility, and strengthen collaborative networks across regions and institutions. In this one-of-a-kind workshop, participants will engage with interactive stations featuring open-capability cryopreservation kits, open-source hardware, and microfluidic platforms. From microfluidics to millifluidics, and cells to organoids, this workshop will stimulate new collaborative opportunities. As such, for 30 days following the workshop, participants will be provided access to our group to develop resin-printed customizable microfluidic and NAM platforms. Together, AGGRC, ZIRC and UAB, will use the workshop to examine the pathway from innovation to implementation.

## Session 7 — Immunity and the Microbiome

### *A canonical neuropeptide pulls double duty as a neuronal marker and an immunodevelopmental cue in the Squid-Vibrio symbiosis*

Briana Tell, Alice Walker, Elizabeth Widun, Leah Columbo, and Elizabeth Heath-Heckman

Michigan State University

Upon birth or hatching animals encounter myriad microbial species which can have wide-reaching effects as they develop. The association between the Hawaiian Bobtail Squid (Euprymna scolopes) and its luminous symbiont (Vibrio fischeri) is one of the most powerful models for bacterial induction of host development. The squid acquires its bacterial symbionts from the surrounding seawater, which then induce apoptosis in and regression of external epithelial appendages. As the signaling mechanisms which govern this process have yet to be fully elucidated, we performed an RNASeq analysis on appendages from animals undergoing this process. Our analysis of the pathways regulated by symbiosis revealed that many of the most highly regulated genes were those involved in neuronal development, guidance, and communication, such as the neurotransmitters serotonin and FMRFamide. As the nervous system of the light organ is yet to be characterized, we identified multiple neuronal cell bodies in the light organ through golgi staining and NeuN antibody labeling, as well as an extensive network of neuronal processes that is present throughout the entirety of the light organ. Through immunocytochemistry we found that many of these neurons are serotonergic and that their processes extend into the blood sinus of the appendages. In addition, we localized the FMRFamide peptide to neuronal processes, but also to the blood sinus inside the appendage, where we confirmed a lumenal increase in staining in response to symbiosis. When localizing FMRFamide transcript, we found that instead of neurons the primary site of production appears to be the circulating immune cells that traffic to the blood sinus in response to symbiosis, providing evidence for a new function for this neuropeptide. Finally, our data show that treatment with FMRFamide induces apoptosis in aposymbiotic animals, likely through the downregulation of nitric oxide production. These data suggest that FMRFamide is a novel, crucial aspect of bacterial-mediated development and the squid-vibrio symbiosis offers a unique opportunity to study the cross-talk between immunity, development, and neurobiology.

## *How the "mimi ika" became a model to study bacterial animal associations*

Michele K. Nishiguchi

University of California, Merced

The mutualistic association between sepiolid squids (Mollusca: Cephalopoda) and their Vibrio symbionts is an experimentally tractable model to study the evolution of animal-bacterial associations through both wild-caught and experimentally evolved populations. Since Vibrio bacteria are environmentally transmitted to new hosts with every generation, it provides a unique opportunity to resolve how changing environmental conditions may affect bacterial infection, colonization, and persistence in an animal model. Vibrio bacteria encounter potentially conflicting selective pressures, competing with one another to colonize the sepiolid light organ, but also vying for resources in the environment outside the squid. Both abiotic and biotic factors contribute to the fitness of individual strains of Vibrio bacteria, but which of these factors are amenable to adaptation and eventually lead to a successful association has yet to be elucidated. Deciphering how environmental conditions and host specificity lead to the development of how Vibrio bacteria can successfully gain the opportunity to infect and spread throughout novel populations has shed light on new ideas on the evolution of bacterial animal associations. This presentation will include several studies both experimentally in the lab and in the field to provide insight into why this model has been a success in uncovering the dynamics of highly specific environmentally transmitted symbioses.

## *Linking host and viral genomics at a wildlife-wastewater interface: a case study from Eagle Bluffs Conservation Area*

Emma E Copen, Caspar Safranski, George Karathanos, Clayton Rushford, Devon Gregory, Ava Barbaro, Matt Frank, Jonathan Naydenov, Marc C Johnson, PhD

University of Missouri–Columbia

Wetlands provide critical ecosystem services, including water filtration, flood mitigation, and habitat for migratory and resident wildlife, making them ecologically important systems to monitor. Among these, wetlands that receive treated wastewater effluent are particularly notable as interfaces where human and wildlife inputs overlap, creating potential points of environmental exposure and zoonotic spillover risk. Eagle Bluffs Conservation

Area, a restored wetland in Columbia, Missouri, exemplifies this type of interface: it is supplied primarily by treated municipal wastewater and lies along the Mississippi Flyway, positioning it as a site where wastewater-associated and migratory wildlife-associated viral signals may intersect. To identify viral lineages and host associations at this interface, we applied a dual genomic discovery pipeline combining viral genomics with vertebrate host screening. Water samples were collected via large-scale concentration and separated into solid and liquid fractions. The solid fraction was used for 12S rRNA metabarcoding to identify vertebrate taxa present at the site. The liquid fraction underwent PEG precipitation to enrich for viral particles prior to RNA metagenomic sequencing. Reads were classified by Kraken2 and GOTTCHA2 for known viral taxa. Reads were also used for de novo assembly and screened by checkV and geNomad for divergent, previously undescribed viral genomes. This approach identified known pathogens infecting species present in our vertebrate candidate list, alongside novel and divergent viral lineages whose disease relevance was uncharacterized. Together, these findings identify candidate host-virus associations at a wildlife-human interface, providing a genomic foundation for future disease discovery and mechanistic study in this understudied system.

## *Novel immune-type receptors as putative natural killer (NK) receptors in ray-finned fish*

Jeffrey A. Yoder

North Carolina State University

Vertebrate genomes encode diverse innate immune receptor families, ranging from ancient lineages to highly derived, species-specific features. While some families are scattered across multiple loci, many reside in genomic clusters formed by tandem gene duplication. Here, I present our research on the Novel Immune-type Receptor (NITR) family in ray-finned fish. Expressed by "NK-like" cells, NITRs are encoded in gene clusters and consist of activating and inhibitory receptors that trigger signaling pathways analogous to mammalian natural killer (NK) cell receptors. NITR genes span all ray-finned fish lineages; their sequence variability and unique genomic architecture indicate rapid, recent evolutionary diversification. Future work aims to map intra-species NITR diversity, identify their ligands, and provide crucial in vivo evidence of NITR immune function.

## Panel Discussion II — New Approach Methodologies

### *Frog on the fly: on-demand Xenopus models for rapid clinical translation*

Richard Novak, Erica Gardner, Bridget Maynard, Viola Fanfani, Rahul Nihalani, Frederic Vigneault

Unravel Biosciences

Unravel Biosciences has built a patient-centric discovery platform using personalized medicine as a means of discovering new therapeutic mechanisms for over 100 genetic disorders. From patient-derived transcriptomic data and our AI algorithms, we construct Living Molecular Twins™ that allow us to model dynamic disease networks and predict drug targets and therapeutic outcomes. To test these predictions, we established SquishyWare™: Xenopus tadpoles as a scalable, cost-effective whole-organism platform for drug discovery in neurodevelopmental disorders.

Tadpoles share physiological, neurological and immune pathways with humans, providing a biologically relevant model enabling much faster and lower-cost studies than organoids or mice. CRISPR generates models (knockout, haploinsufficiency, mosaicism) in 3 weeks, while mRNA co-expression is used to test the effects of variants (gain-of-function, dominant-negative, neomorph) and mimic gene therapy. Squishyware™ motion-tracking assesses real-time behaviors in 100-200 tadpoles simultaneously, bridging AI drug predictions with in vivo outcomes.

This closed-loop patient RNA to phenotype and back to patient framework nominated RVL-001 (vorinostat liquid oral formulation), demonstrating whole-organism rescue in tadpoles and mice, securing Orphan Drug Designation, and now in a clinical trial for Rett Syndrome. Additionally, SquishyWare™ models were used to derisk a novel gene therapy as part of an IND clinical trial submission package and recapitulated a highly heterogeneous patient population. This talk will highlight the value of Xenopus in enabling validation of highly effective, low-toxicity molecules as a practical step before more expensive and time-consuming models and clinical implementation.

## *Closing the translational gap: Xenopus advances clinical development in Rett syndrome*

Gardner, Erica; Maynard, Bridget; Fanfani, Viola; Quach, Linda; Reitsma Mikayla; Pitsiniaga, Eleni; Flores, Romano; Nihalani Rahul; Muni, Neal; Vigneault, Frederic; Novak, Richard

Unravel Biosciences

Rare and complex disorders remain difficult to translate from molecular hypothesis to clinical benefit. Unravel Biosciences, a clinical-stage therapeutics company, has developed a patient-centric platform applied across more than 100 disorders that integrates patient transcriptomics, Living Molecular Twins™ (LMTs), BioNAV™ computational drug prioritization, and rapid whole-organism testing in Xenopus laevis to reduce uncertainty earlier in development.
Patient RNA is used to construct dynamic disease-network models and prioritize compounds predicted to normalize dysregulated pathways. CRISPR-enabled Xenopus models can be generated within weeks, providing an intact vertebrate system for evaluating functional efficacy, dose response, tolerability, and multisystem biology. In Rett syndrome, tadpoles recapitulated developmental, seizure-like, motor, and behavioral abnormalities, while scalable machine-vision phenotyping enabled quantitative in vivo screening at clinically relevant doses.
This integrated workflow identified vorinostat as a lead candidate. Xenopus studies demonstrated whole-organism rescue and contributed to candidate prioritization as part of a broader preclinical evidence package. Subsequent studies in symptomatic MeCP2-deficient mice showed broad CNS and non-CNS efficacy when treatment was initiated after disease onset, strengthening translational confidence.
The combined evidence supported advancement of vorinostat as RVL-001 into the ongoing RETT REVOLUTION clinical study in Colombia; FDA Orphan Drug Designation for Rett syndrome was granted in May 2024. This program illustrates how Xenopus can accelerate development and reduce translational risk by testing patient-derived therapeutic hypotheses in a living vertebrate before greater investment in mammalian studies and clinical trials.

## Session 8 — Toxicology, Environmental Exposure, and Disease Risk

### *Persistent behavioral changes following parental benzo[a]pyrene exposure in Japanese medaka (Oryzias latipes)*

Nonnie E Cook, Yuan Lu, Hunter Kulesz, Rijith Jayarajan, Riccardo Mozzachiodi, Frauke Seemann

Texas A&M University-Corpus Christi

Benzo[a]pyrene (BaP), a ubiquitous polycyclic aromatic hydrocarbon, is a persistent environmental contaminant linked to developmental toxicity, yet its long-term behavioral consequences across generations remain insufficiently characterized. This study tested whether parental BaP exposure induces anxiety-like and locomotor behavioral alterations detectable in unexposed adult offspring and whether these phenotypes are associated with cell type-specific disruption of brain transcriptional programs. Adult medaka (Oryzias latipes) were exposed to either solvent control or 1 µg/L BaP for 21 days. Following exposure, eggs were collected and F1 offspring were reared under clean conditions to 6 months of age. Adult offspring were evaluated using behavioral assays of locomotor activity, vertical swimming, and light/dark preference, and brain tissue from adult males was profiled using single-nucleus RNA sequencing (snRNA-seq).

BaP-lineage offspring exhibited more frequent light–dark transitions and increased entries into the lower portion of the tank. BaP-lineage males showed enhanced dark preference and reduced freezing duration, indicating a stress-responsive phenotype. Single-nucleus transcriptomic profiling resolved 14 major brain cell populations and identified cell type-specific responses to parental exposure. The strongest transcriptional alterations occurred in midbrain glutamatergic neurons, cerebellar Purkinje neurons, pineal gland cells, and GABAergic interneurons. Pathway enrichment analyses highlighted alterations related to protein and cytoskeletal binding, nucleotide binding, ion transport and channel activity, enzyme binding, and lipid binding.

These findings show that parental exposure to BaP can alter behavior in adult male offspring and selectively affect gene expression across neural cell populations, linking cross-generational behavioral phenotypes with cell-resolved molecular changes in the vertebrate brain.

## *Discovery of the natural substrates of the drug transporter P-glycoprotein*

Evan Tjeerdema, Sanjoy Adak, Alex Babcyzk, Lisa Hueck, Rachel Metry, Sascha Nicklisch, Brad Moore, Amro Hamdoun

Scripps Institution of Oceanography

The transporter P-glycoprotein (P-gp/ABCB1) is well studied for its role in drug disposition. It is expressed at barrier sites where it effluxes toxicants, thereby protecting against accumulation and toxicity. Yet, despite half a century of research on the role of this protein in transport of man-made drugs and pollutants, the naturally-occurring molecules transported by P-gp remain largely unknown. Based on our prior work with persistent pollutants, we hypothesized that P-gp might transport organohalogens produced by marine bacteria (MOHs). To probe this question, we developed a high-throughput screening pipeline leveraging automated dosing, imaging, and analysis of P-gp-knockout sea urchin embryos. We screened a library of 68 MOHs sourced from Streptomyces, Pseudoalteromonas, Hormoscillia spongilae, and Aetokthonos hydrillicola. Of these, 16 compounds exhibited different toxicity profiles between controls and P-gp mutants, with control EC50 values of 0.10-5.60 μM and 0.017-3.79 μM for knockouts, corresponding to a 1.34-5.86-fold increase in toxicity of these compounds to P-gp-knockouts. Tetrabromopyrrole and four monodeoxypyoluteorins showed the greatest toxicity increase to P-gp-knockouts. Analysis of their differential accumulation patterns using LC-MS revealed that P-gp-knockouts embryos accumulated 1.25-2.00X more of these compounds than controls. To test whether these interactions might be conserved across species, we used ATPase assays of human P-gp and found that three of these MOHs were substrates of human P-gp with EC50 values of 5.29-58.6 μM. This discovery suggests that organohalogen marine natural products might drive drug transporter evolution, differentially accumulate across species, and interact with pollutants to enhance toxicity through their action as substrates of P-gp.

## *Uncovering the role of Bmp-chemokine signaling in ethanol-induced cranial neural crest apoptosis*

Makenzie C. Stearsman, CB Lovely

University of Louisville

Fetal Alcohol Spectrum Disorders (FASD) affect up to 5% of US school-aged children, making it a crucial health problem. While facial phenotypes are

common in FASD, the genetic risk factors and their impacted cellular mechanisms are poorly understood. Endoderm and cranial neural crest cell (CNCC) signaling are critical for facial development. CNCC migrate into the pharyngeal arches, where signals from the endoderm direct facial cartilage development. We have previously shown that Bone Morphogenetic Protein (Bmp) signaling is ethanol-sensitive and required for endoderm and CNCC development. Ethanol does not impact Bmp signaling directly but increases apoptosis in migrating CNCC in Bmp mutants. This suggests Bmp signaling regulates migrating CNCC through an unknown mechanism. Chemokine signaling through cxcl12a/b and cxcr4a/b regulates CNCC migration, and its disruption causes CNCC apoptosis and craniofacial defects. Previous work links Bmp and chemokine signaling in cell migration, suggesting chemokine signaling as the Bmp-dependent signal to migrating CNCC. We utilize zebrafish due to their genetic conservation and tractability, external fertilization, and large numbers of translucent embryos. Hybridization Chain Reaction (HCR) Fluorescent in situ Hybridization (FISH) shows that cxcl12a/b are expressed in the endoderm, while cxcr4a/b are expressed in migrating CNCC. Bmp inhibition with dorsomorphin (DM) disrupts cxcl12b expression in the endoderm, but not cxc4ra/b in the CNCC. Overall, this project will provide greater insight into the complex, ethanol-sensitive signaling events leading to craniofacial defects in FASD. This work will directly translate to studies of candidate genes in human populations and allow for better diagnosis and treatment of FASD.

## *Effects of chronic nitrite exposure on maternal health and reproduction during gestation in Xiphophorus couchianus*

Trinity Parkman, Yuan Lu, and Mar Huertas

Department of Biology, Texas State University

Nitrogenous fertilizers are widely used in agriculture; however, fertilizer runoff leads to heavy accumulation of nitrite and nitrate in aquatic environments. While acute nitrite exposure has been shown to have multiple systemic effects, its effects on reproduction and its chronic effects remain understudied. To investigate this, we used the live-bearing fish Xiphophorus couchianus to elucidate the effects of chronic nitrite exposure throughout pregnancy on maternal health and reproduction. Following exposure to 0 mM, 0.01 mM, 0.1 mM, and 1 mM nitrite concentrations in water for the 30-day gestation period of X. couchianus, we quantified embryonic stages, analyzed morphological changes in the ovary, and extracted RNA from maternal head kidneys. We found a significant decrease in the number of eggs and embryos

produced in the higher-treatment groups, suggesting that chronic nitrite exposure may impair fertilization and interfere with embryogenesis. Ovarian histology in the 1 mM nitrite treatment, compared with controls, showed a significant increase in atresia and immune infiltration. Transcriptome analysis of the head kidneys revealed a significant decrease in immunity at the 0.01 and 0.1 mM nitrite treatments. In contrast, the 1 mM nitrite treatment shifted from immunosuppression toward upregulated stress responses, notably DNA damage and repair pathways. Overall, chronic nitrite exposure during pregnancy reduces female fecundity and significantly suppresses maternal immunity, particularly at lower concentrations, even without external signs of stress (e.g., loss of appetite or erratic movement). Therefore, aquaculture and drinking water should routinely monitor nitrite levels, especially where exposure may affect pregnant individuals.

## *The killifish model for revealing mechanisms that underlie resilience to cardiotoxic and neurotoxic pollutants*

Andrew Whitehead, Nicole McNabb-Kelada, Lars Anderson

University of California, Davis

Fundulus killifish are a powerful model system for studying genotype-by-environment interactions for exposure-induced disease. This is because multiple populations of two species (Fundulus heteroclitus and Fundulus grandis) have rapidly and repeatedly evolved adaptations to persist in urban estuaries that are highly modified, including by pollution with complex mixtures of toxic chemicals. Natural selection has therefore amplified the frequency of otherwise rare genetic variants that affect susceptibility to exposure-induced disease. Molecular, population genomic, and quantitative genetic studies within and among killifish populations have provided insights, at the physiological, developmental, molecular, and genetic levels, into the mechanisms that enable persistence in these extreme environments, including the molecular pathways, genes, and variants that modulate disease susceptibility. Much of this work has focused on adaptations to chemicals that cause cardiovascular developmental disease. But neurotoxic chemicals are the most abundant pollutants in these urban sites, where urban killifish encounter toxic concentrations. Pilot studies show that early-life exposure to neurotoxic PCBs affect larval behavior in non-urban killifish, but not in urban killifish, providing the first evidence of evolved resistance to exposure-induced neurological disease in killifish. Ongoing studies seek to extend the killifish model system to accelerate discovery of the genetic and neurobiological mechanisms that affect behavioral resilience to common pollutants.

## Session 9 — Behavior, Neuroendocrinology, and Physiology

### *From gravity sensation to balance dysfunction*

Yunlu Zhu

University of Michigan

The sensation of gravity anchors our perception of the environment and provides a fundamental reference for balance. When this system is disrupted, balance impairments manifest as postural instability and gait abnormalities, hallmarks of many neurological disorders. However, how neural circuits transform gravitational information into motor commands to maintain balance remains poorly understood. Using larval zebrafish, we combine quantitative behavior, functional imaging, and circuit manipulation to define how animals maintain balance, how environmental and circadian signals shape locomotor strategies, and how brainstem accumulation of pathogenic protein tau leads to balance deficits.

### *Orphan cytochrome P450 20a1 mutants and behavioral phenotypes in zebrafish*

Jared V. Goldstone, Nadja R. Brun, Matthew C. Salanga, Maya Cohen, Maeve Farley, Francisco X. Mora-Zamorano, David C. Lamb and John J. Stegeman

Woods Hole Oceanographic Institution

Cytochrome P450 20A1 (CYP20A1) is the last human P450 enzyme without an identified substrate, and orthologs occur as single genes in every vertebrate genome sequenced to date. The occurrence of high levels of CYP20A1 transcripts in human substantia nigra and hippocampus and abundant maternal transcripts in zebrafish eggs strongly suggest roles both in the brain and during early embryonic development. Homozygous zebrafish cyp20a1 null mutants exhibit significant behavioral differences from wild-type zebrafish, both in larval and adult animals. Larval cyp20a1-/- mutants exhibited a strong increase in light-simulated movement (i.e., light-dark assay). Further, the larvae exhibited mild hypoactivity during the adaptation period of the optomotor assays. Adult cyp20a1 null fish showed a pronounced delay in adapting to new environments, which is consistent with an anxiety paradigm. RNA-seq data show that an array of genes, including kcnh3, kcnk10b, gabar4, slc6a6a are strongly downregulated in both larvae and adult brains of CYP20 null zebrafish. nNon-targeted metabolomics has identified some metabolite

differences that may contribute to or underly impaired neuronal inhibition and synaptic homeostasis. Lower calcitriol, palmitoleic acid, and dihydropteroate levels are observed in the Cyp20a1 knockout lines. These lines of evidence point to CYP20A1 playing an important role in the developing and adult nervous system, across vertebrates.

### *Identifying the genetic factors underlying reduced stress response in Astyanax mexicanus*

Stefan Choy, Rianna Ambosie, Maya Enriquez, Emilie Richards, Aubrey Manning, Alexander Seaver, Suzanne McGaugh, Erik Duboue, Johanna Kowalko

Lehigh University

After experiencing a stressor, animals experience distinct physiological and behavioral responses. The genetic bases for stress responses are evolutionarily deeply conserved, though how stress response is genetically modulated is not well understood. To investigate the genetic basis of stress response we utilize Astyanax mexicanus, a freshwater fish with multiple, interfertile populations of riverine surface fish and over 35 populations of cavefish. A. mexicanus cavefish have repeatedly evolved multiple traits, including eye loss, albinism, and a reduced stress response. We quantified multiple traits, including stress response, in cave-surface F2 hybrids derived from multiple cavefish populations. We found that several traits co-vary with stress response in cave-surface hybrids, including albinism. We directly tested for pleiotropy by measuring stress response in surface fish with CRISPR-Cas9 induced mutations in oca2, the gene responsible for albinism in cavefish. Albino surface fish have a reduced stress response, suggesting that oca2 plays a role in the evolution of reduced stress response in A. mexicanus. Next, we performed quantitative trait loci mapping and identified genomic regions that overlap for distinct traits, suggesting a shared genetic basis in the evolution of traits in multiple cavefish populations. My current project is identifying genes that are differentially expressed between cave and surface fish after experiencing a stressor. We will combine these results with the QTL data to generate a strong candidate list for genes underlying the evolution of reduced stress response in A. mexicanus. Together, these experiments increase our understanding of how stress response evolves and which targets evolution enacts upon.

## *Neural and behavioral signatures of environmental stress in zebrafish (Danio rerio)*

Rachael M. Heuer, Jonathan Cordle, David Ottervall, Katie Hastings

University of Miami Rosenstiel School of Marine, Atmospheric, and Earth Science

Extreme heat events and environmental pollutants are intensifying human exposures, and both are increasingly recognized as threats to neurological, cognitive, and developmental health. Aquatic animal models offer a powerful and accessible way to study how these stressors act on the developing nervous system. In this study, we used larval zebrafish (Danio rerio) to integrate whole-brain activity mapping with high-throughput behavioral assays under environmental stress. Using MAP-mapping, a phosphorylated-ERK-based approach for whole-brain activity mapping, we compared control and heat-exposed animals and found that heat-exposed zebrafish show differential activation across distinct brain regions, alongside altered behavior including a suppressed startle response and faster habituation. We also find that polycyclic aromatic hydrocarbons (PAHs), which are both neurotoxic and carcinogenic, alter the startle response both as individual compounds and in complex mixtures. Because stressors such as heat and PAHs frequently overlap in environmentally relevant settings, a central strength of this system is the ability to test how multiple co-occurring stressors interact to affect the nervous system. We are extending this approach to combined heat and pollutant exposures, as well as to other pollutants of concern such as phthalates, to assess how human-relevant contaminant mixtures affect neural activity and behavior. By linking brain-wide neural activity with quantitative behavioral measures within a single tractable vertebrate system, zebrafish provide a powerful platform for identifying neural signatures of environmental exposure and investigating how climate-related and chemical stressors may contribute to neurological dysfunction.

## *Zebrafish screen of schizophrenia risk genes reveals convergent dysregulation of cholesterol metabolism*

Summer Thyme

UMass Chan Medical School

Rare coding variants provide a tractable entry point for understanding the molecular mechanisms underlying schizophrenia risk. Here, we generated and characterized zebrafish lines with mutations in the orthologs of >20 human

schizophrenia-associated genes, including eight of the top ten SCHEMA genes, genes disrupted in childhood-onset schizophrenia (COS), and genes located within recurrent copy number variants. Whole-brain phospho-Erk activity mapping and behavioral profiling identified phenotypes in multiple mutant lines. We prioritized a protein-truncating mutation in sp4, which encodes an activity-dependent transcription factor, and a COS-associated missense mutation in atp1a3a, which encodes a Na+/K+ ATPase pump, for additional characterization. Both knockout and point mutations in atp1a3a disrupted brain activity and behavior in larvae and impaired navigation of a Y-maze in juveniles. Bulk RNA sequencing data from adult sp4 and atp1a3a brains highlighted convergent upregulation of sterol biosynthesis pathways, including increased expression of srebf2 and msmo1. Analysis of previously published telencephalon single-cell data demonstrated that cholesterol synthesis genes are enriched in astrocyte-like cells and increase in expression during post-larval development. Consistent with transcriptomic findings, filipin staining indicated increased free cholesterol in juvenile sp4 and atp1a3a mutant brains. Our findings identify dysregulation of glial and sterol-associated programs as a shared molecular consequence of two distinct schizophrenia risk mutations. Although whether sterol pathway dysregulation represents a primary pathogenic mechanism or a secondary response to changes in neuronal activity requires further investigation, the convergence observed between genetic models and developmental stages suggests that disruptions to lipid homeostasis could represent a shared feature of schizophrenia disease biology.

## Session 10 — Evolutionary and Comparative Disease Models

### *GeroEvoDevo: deep homology of the regulation of longevity and aging across metazoan…from ancient leviathans to humans*

Matthew Harris

Harvard Medical School

Aging is the primary cause of morbidity and disease that affects our society. Understanding how to alleviate aging will have profound effects on efforts to increase heath and lifespan. Longevity, apart from aging, is an evolutionary trait that is specific and defining for particular lineages, and thus heritable and shaped by evolution. Studying how longevity is mediated in cases of extreme lifespan uncovers Nature's unique strategies to staved off aging and maintain homeostasis for extended periods. We use a phylogenetic approach to look at comparative genetics underlying exceptional longevity in Nature. We capitalize on replicative cases of varied lifespans within clades to hone genomic signals across phylogenies that result in Evolutionary Gene Sets (EGS) of longevity. In these cases, we apply these EGS against variation data in the 1-5% oldest humans within a population cohort to refine GWAS analysis. We apply a phylogenomic analysis to Lifespan evolution in Rockfishes and more broadly to venerid clams. An unexpected synergy between all these broadly related metazoans in the regulation of longevity suggests such Gerontological Evolutionary Development (GeroEvoDevo) approaches will provide the lens to bring such complex phenotypic life history traits into focus.

### *Poison frogs as an emerging model in human disease, pharmacology, and behavior*

Roberto Márquez

Virginia Tech

Over the past decades it has become clear that evolutionary approaches can be fruitful to understand the molecular mechanisms underlying human diseases and their treatments. The vast majority of our knowledge of these molecular mechanisms is derived from work on a handful of species, often called the "model" organisms. Similarly, a good amount of the work informing our understanding of ecological and evolutionary processes has focused on a handful of well-studied lineages and ecosystems. Interestingly, there isn't

much overlap between these two groups of model systems, which hinders our ability to understand how the transfer of molecular information across generations can give rise to particular phenotypes. In this talk I will share my and my colleagues' progress to establish poison frogs as a model for integrative studies that bridge the gap between molecular mechanisms and the evolutionary processes that give rise to them, allowing us to leverage these animals' unique biology towards a deeper understanding of human disease.

## *Origins of aryl hydrocarbon receptor ligand specificity*

Tessa Steiniche, Sibel I. Karchner, Jared V. Goldstone, Chesna Mandl, Joseph W. Thornton, Mark E. Hahn

Woods Hole Oceanographic Institution

The aryl hydrocarbon receptor (AhR) is an evolutionarily conserved transcription factor with pleiotropic roles in xenobiotic metabolism, immunity, and development. Vertebrate AhRs are ligand-activated and notably promiscuous, binding a structurally diverse array of endogenous and xenobiotic compounds. Whether invertebrate AhRs are similarly ligand-dependent, and whether ligand-dependence is an ancestral or derived feature of the AhR, remains largely untested. To address this, we assayed AhRs from invertebrate taxa representing key phylogenetic transitions: pre-bilaterians (Cnidaria—N. vectensis), protostomes (Nematoda—C. elegans; Arthropoda—D. melanogaster; Mollusca—M. arenaria), deuterostomes (Echinodermata—S. purpuratus), and invertebrate chordates (Urochordata—C. intestinalis; Cephalochordata—B. floridae). Chimeric constructs with species-specific ligand-binding domains (LBDs) were expressed in a one-hybrid reporter system and screened against more than 20 prototypical AhR ligands. With the exception of invertebrate chordates, no invertebrate AhR responded to ligand treatment. Protostome AhRs displayed robust ligand-independent transcriptional activity, suggesting that constitutive activation is a derived feature of the protostome lineage. Several control experiments were performed to rule out experimental artifacts. To probe the mechanistic basis of constitutive activity, we mutated the M. arenaria LBD and identified two adjacent residues, an alanine and a tyrosine that together shift local polarity and fill the binding cavity, as responsible for nearly all constitutive activity. Functional analyses of extant invertebrate and inferred ancestral AhRs are ongoing. By tracing the emergence and diversification of AhR ligand-binding properties, this work offers new insight into the early evolutionary history of AhR signaling. [Research supported by NIEHS: R01ES035261.]

## *Evolutionary insights from a brain in darkness: Astyanax mexicanus as a model of resilience*

Rachel A. Carroll, Edward S. Ricemeyer Katie Gallman, Maggs X, Robert Peuß, Nicolas Rohner, Johanna Kowalko, Suzanne McGaugh, Alex C. Keene, Wesley C. Warren

University of Missouri

The Mexican tetra, Astyanax mexicanus, has phenotypically distinct cave- and surface-dwelling morphs, providing a naturally evolved model for investigating brain adaptation to complete darkness, food scarcity, and hypoxia. Several behavioral phenotypes differ between morphs, most notably sleep loss in cavefish. Using single-nucleus RNA sequencing, we classified 117,000 nuclei from adult cave and surface fish brains and identified 35 neuronal and non-neuronal cell types. Bayesian proportionality analysis revealed significant compositional differences: oligodendrocytes and VGLUT1+/VGLUT2b+ neurons were enriched in surface fish, whereas cavefish had higher proportions of cerebellar ventricular-zone neurogenic progenitors, Purkinje cells, immune cells, VGLUT1+ Eury dendroid neurons, and unlocalized VGLUT1+ neurons. Pairwise comparisons identified differentially expressed genes (DEGs) across all cell types associated with sex, natural sample variation, and ecotype, with ecotype being the strongest driver of within-cell-type variation among immune and neuronal populations. Bulk RNA sequencing identified 5,891 significant DEGs. Cross-referencing the top 200 DEGs with weighted gene co-expression network analysis highlighted cavefish-associated genes, including her9, which is involved in photoreceptor formation and retinal organization, and phf20a, an epigenetic regulator linked to development and stress adaptation. Together, these findings reveal cellular and gene-regulatory changes that may support cavefish adaptation and resilience to dark, resource-limited environments.

## *Thyroid hormone signaling drives differences in developmental shape covariation across the flatfish skull.*

Kory Evans

Rice University

Developmental integration emerges from coordinated interactions among traits, tissues, and genes, while modularity allows anatomical regions to vary with partial independence. Although developmental integration has been proposed to follow a palimpsest model in which successive processes

overwrite earlier patterns of covariation, it remains unclear how integration is reorganized during major developmental transitions or how endocrine signals coordinate this process across morphology and gene expression. Flatfish metamorphosis provides a powerful system to address these questions because larvae undergo dramatic cranial remodeling as they transform from bilaterally symmetrical pelagic larvae into asymmetrical benthic juveniles. Thyroid hormone (TH) is a central regulator of this transition, and disrupting TH signaling can arrest metamorphosis. I hypothesize that flatfish metamorphosis involves TH-dependent rewiring of cranial integration, reorganizing both morphological modules and gene expression programs. I will test this hypothesis through three objectives. First, I will use three-dimensional geometric morphometrics to determine how skull integration and modularity change during southern flounder metamorphosis. Second, I will compare TH-inhibited larvae with controls to test whether endocrine perturbation alters the strength or anatomical organization of developmental covariance. Third, I will use single-cell RNA-seq to identify co-expression modules associated with metamorphosis, craniofacial remodeling, and endocrine response, and test whether TH alters transcriptional covariance among developmental programs. By integrating morphology, experimental perturbation, and transcriptomics, this project will establish flatfish metamorphosis as a model for understanding how endocrine signaling dynamically reorganizes developmental systems to generate coordinated morphological change and anatomical complexity. These results will reveal general principles linking hormonal regulation, integration, and modularity.

## Session 11 — Nutrition and Chronic Disease

### *Vitamin D and nutrient sensing in zebrafish*

Wolfram Goessling

Yale School of Medicine

Zebrafish are ideally suited to interrogate the impact of developmental stressor on metabolic health in adulthood. We have developed high-fat diet feeding protocols to induce body mass increases and liver steatosis in larvae and adult zebrafish. Population studies have demonstrated a correlation between vitamin D deficiency and steatotic liver disease, but the exact role of vitamin D receptor signaling in nutrient sensing and lipid metabolism has not been elucidate. Here, we illuminate the evolutionary function of VDR by demonstrating that zebrafish Vdr coordinates hepatic and organismal energy homeostasis through antagonistic regulation of nutrient storage and tissue growth. Hepatocyte-specific Vdr impairment increases hepatic lipid storage, partially through acsl4a induction, while simultaneously diminishing fatty acid oxidation and liver growth. Importantly, Vdr impairment exacerbates the starvation-induced hepatic storage of systemic fatty acids, indicating that loss of Vdr signaling elicits hepatocellular energy deficiency. Strikingly, hepatocyte Vdr impairment diminishes diet-induced systemic growth while increasing hepatic and visceral fat in adult fish, revealing that hepatic Vdr signaling is required for complete adaptation to food availability. Our findings establish hepatocyte Vdr as a regulator of organismal energy expenditure and define an evolutionary function for VDR as a transcriptional effector of environmental nutrient supply.

### *Establishing the freshwater crustacean, Daphnia pulex, as an alternative, short-lived animal model of healthspan and aging: impacts of diet and culture medium*

Victoria K. Gibbs, Yuna Kim, Michelle Zheng, Rebecca L. Inglesias, and Gustavo Roveri

University of Alabama at Birmingham

Daphnia pulex is commonly used in ecotoxicology and has potential as a model for aging biology due to its parthenogenic reproduction, short lifespan (<60 days), and increased lifespan with dietary restriction. Culture conditions vary across laboratories using D. pulex for longevity studies, thus generalizations of lifespan interventions are difficult. We compared published

culture conditions to discern effects on health and longevity in D. pulex. In Experiment 1, we fed daphnia live, unicellular green algae, Raphidocelis subcapitata (GA) and compared culture media, using either reconstituted moderately hard freshwater described by the Environmental Protection Agency (EPA) or COMBO medium formulated with trace elements. In Experiment 2, we used COMBO medium and fed daphnia either GA or a commercial, non-living algae concentrate, Shellfish Diet (SD). Twenty-four neonates derived from four clonal lines were housed individually in glass vials containing 50 mL of the respective medium. Individuals were fed three days per week and inspected daily for survival and reproduction. Body length was measured weekly. The median lifespan (35.5±16 and 36.5±13 days) and average body length (3.10±0.16 and 3.09±0.12 mm) were similar for individuals held in either EPA or COMBO culture media; however, average total offspring produced was highest for those held in COMBO medium. The median lifespan was highest for individuals fed SD (74.5±22 and 38±13 days for SD and GA, respectively). Under the conditions of this study, culture medium impacted reproduction, and diet influenced lifespan. We will further investigate diet to determine effects of dietary restriction or nutritional composition on lifespan.

## *Epigenetic encoding of hypoxia resistance in the sea hare Aplysia californica*

Javier A. Rodriguez-Casariego, Jose M. Eirin-Lopez, Mark W. Miller, Michael Schmale, Lynne A. Fieber

Florida International University

Accumulating evidence suggests that epigenetic mechanisms are critical in the establishment and inheritance of “cellular memory”, a process highly relevant to human disease susceptibility and neuroprotection. However, it is unclear how environmental-exposure-driven epigenetic modifications could affect short- and long-term gene expression responses, nor the specifics of its mitotic and meiotic inheritance. We have developed short and long-term hypoxia exposure protocols in the neurobiology model mollusk Aplysia californica, showing that hypoxia resistance in this species is facilitated by environmental memory acquired after exposure. This offers a unique, translationally relevant system to study conserved molecular mechanisms of neuroprotection under hypoxic-ischemic conditions like stroke. Expanding on this work, here we characterized changes in base-specific DNA methylation (WGBS) in neurons of the abdominal ganglion from individuals with different generational hypoxia exposure histories. Consistent with other marine

invertebrates, the Aplysia methylome is predominantly genic and positively correlated with baseline gene expression. Crucially, we identified an 'epigenetic priming' signature with lineage-specific differences in DNA methylation at neuroprotective and signaling loci. Under hypoxia, the two cohorts exhibited divergent temporal programs with large methylation shifts associated with gene expression regulation. Integrative network analysis revealed that while both cohorts remodel shared, highly conserved hypoxia-response hubs including HIF1A and SMAD3, pre-conditioned animals selectively remodel metabolic and comprehensive DNA repair modules, whereas naïve animals prioritize ubiquitin-mediated proteostasis. These results demonstrate that environmental history is encoded through DNA methylation to stabilize protective states, providing an initial mechanistic framework for the transgenerational inheritance of hypoxia resistance in the nervous system.

## *Expanded site-directed transgenesis tools for zebrafish and beyond*

Seth Jacobson, Swapnil Hingole, J. Michael Swing, Emma Harrison, Rose Doss, Jack Ustanik, Amanda Garfield, Mimi Morrison, Martin Breuss, Alexa Burger, Christian Mosimann

University of Colorado Anschutz Medical Campus

Standard transgenesis across model systems relies on random genomic integration, requiring extensive screening and validation. In contrast, targeted integration into pre-validated genomic safe harbor loci using phiC31 integrase-mediated attP/attB recombination has transformed transgenesis in mouse and Drosophila with predictable, reproducible transgenesis. Termed phiC31 integrase Genomic Loci Engineered for Transgenesis (pIGLET), we previously established two safe harbor-type attP landing sites in zebrafish to support numerous transgene applications. While broadly applicable, the two landing sites pIGLET14a and pIGLET24b limit combinatorial transgene applications to two loci in the genome.
In addition to streamlining pIGLET-based applications and protocols for community access, we are now expanding the phiC31 integrase-based transgenesis toolkit by generating additional safe harbor landing sites across the zebrafish genome. Our strategy combines MinION long-read sequencing to map robust Tol2 transgene insertions with CRISPR-Cas9-mediated knock-in of attP sites. Our workflow has identified several new candidate safe harbor loci, including potentially methylation-resistant regions suitable for repetitive transgenes. Comparative analyses across aquatic model systems indicate that

our identified genomic loci are potentially available for phiC31 integrase-mediated transgenesis beyond zebrafish. We are further establishing quasi-enhancer trap lines by inserting attP landing sites into the 5′ UTRs of genes with distinct expression patterns, enabling targeted integration of fluorescent reporters with previously inaccessible expression patterns: we piloted this approach with the hematopoietic stem cell marker runx1 and the canonical Wnt target axin2. Together, our efforts establish and expand a versatile platform for efficient, reproducible, and accessible targeted transgenesis in biomedical research using zebrafish and other aquatic models.

## *Comparative nutrition in fish: applications to human health and agricultural production*

Biga, P.R., M. Addo, and M.N. Latimer

Texas State University

Fish are widely used as biological models for investigating nutrition, metabolism, and disease, while also serving as important food sources that contribute essential nutrients, including omega-3 fatty acids, to human diets. Consequently, nutrition research in fish has broad implications for both human health and agricultural production. Methionine, an essential amino acid, plays a central role in protein synthesis, one-carbon metabolism, and epigenetic regulation through its function as a methyl donor. Although methionine restriction is shown to improve metabolic health and extend lifespan across diverse animal models, methionine supplementation during critical developmental windows can enhance growth and production in aquaculture species. Our research demonstrates that methionine availability regulates muscle growth through multiple molecular mechanisms. In rainbow trout, methionine influences the expression of genes associated with muscle development via DNA methylation and microRNA-mediated regulation. Specifically, methionine availability alters the expression of microRNAs, including miR-133a and miR-210, as well as genes involved in muscle growth and metabolism, such as Bmal1, Clock, and Ror-alpha. We further show that methionine availability modifies DNA methylation patterns associated with circadian rhythm, oxidative stress, translational regulation, and apoptosis. In addition, parental methionine intake has significant effects on offspring programming and gene expression. These findings highlight methionine as a key nutritional regulator linking dietary inputs to epigenetic mechanisms that influence growth and development. Collectively, this work advances our understanding of nutritional physiology and demonstrates how fish models provide valuable insights applicable to improving aquaculture production

while informing nutritional strategies relevant to human health and plant-based feed development.

## Keynote Lecture 3

### *The Mexican tetra Astyanax: a model for the biology of heart asymmetry*

William R. Jeffery

University of Maryland

Vertebrates have asymmetric hearts positioned with a bias toward the left side of midline. Alterations in left-right (L-R) asymmetry can lead to cardiac malfunctions, but the underlying mechanisms are poorly understood. Natural L-R cardiac reversal (situs inversus) has been difficult to study because of its extreme rarity in most vertebrate species, including humans. The Mexican tetra, Astyanax mexicanus, consists surface-dwelling (surface fish, SF) and cave-dwelling (cavefish, CF) morphs. In addition to loss of vision and pigmentation, CF show up to 50% reversal of normal heart asymmetry. Thus, the Mexican tetra is an excellent model to study defects in heart asymmetry. The situs inversus phenotype is inherited in successive CF generations, suggesting a genetic basis. SF and CF are interfertile, and reciprocal hybridizations reveal a maternal basis for CF situs inversus, focusing our efforts on early embryogenesis. Accordingly, we find that Kupffer's Vesicle (KV) (the teleost L-R organizer) development and the downstream Dand5-Nodal signal are modified in CF. Increased Sonic hedgehog (Shh) signaling along the CF midline is instrumental in this modification. The natural cave environment is hypoxic, and SF exposed to hypoxia in the laboratory show increased Shh expression and corresponding changes in the KV and Dand5-Nodal system resembling the natural CF phenotype, indicating that hypoxia is a major inducer of situ inversus. CF with left and right oriented hearts thrive in both natural and laboratory environments and will be suitable subjects in future examinations of the relationship between L-R asymmetry and cardiac health.

# Poster Session I

## *Zebrafish 3D microanatomical atlas: a reference framework for multimodal imaging and 3D phenotyping*

Khai C Ang, Alex Y Lin, Jean E Copper, Mee S Ngu, Spencer Katz Rachelle Saint-Fort, Skye Bixler, Mehmet S. Aslan, Ahmed Shokry, Daniel J Vanselow, & Keith C Cheng

Penn State College of Medicine

The zebrafish 3D microanatomical atlas is a whole-organism reference resource that integrates high-resolution, isotropic microCT imaging with ontology-based vector annotations and multimodal omics data within a

architecture, the atlas addresses major limitations of conventional 2D histology and retains anatomical relationships across organs, tissues, and developmental stages. Its common coordinate framework will provide an anatomical backbone for registering complementary imaging modalities, including histology, fluorescence microscopy, and ultrastructural microscopy, within the same 3D reference space.

Integration with spatial transcriptomics and other omics technologies extends the atlas from an anatomical reference to a platform for linking molecular profiles with cellular and tissue features throughout the organism. This framework supports systematic, organism-wide phenotyping of genetic, toxicologic, and environmental perturbations while maintaining their spatial context. Standardized anatomical annotations further support reproducible comparisons across specimens, experiments, developmental stages, and data modalities.

Its digital architecture and standardized annotations are designed for AI readiness, facilitating machine learning-based computational phenotyping, cross-modality data fusion, and predictive modeling that link zebrafish perturbations to human disease mechanisms. By providing open, harmonized multimodal datasets within a shared coordinate system, the atlas establishes a community resource for cross-species comparisons, translational research, and the development of quantitative 3D vertebrate pathology.

## *Engineering quasi-enhancer traps for endogenous expression-driven transgenes in zebrafish*

Swapnil Hingole, Seth Jacobson, Raéden Gray, Marycruz Flores Flores, Mimi Morrison, Alexa Burger, Christian Mosimann

University of Colorado Anschutz

Transgenic zebrafish are key reagents for comparative medicine studies, building on the versatile genetic tools available in the model. Transgenes provide expression of transgenic cargo including fluorophores, effectors, and recombinases such as Cre in cell types and processes of interest through characterized regulatory elements. However, suitable regulatory elements for numerous cell types and processes have remained unavailable. In addition, transgenics generated from isolated and confined regulatory elements at times incompletely recapitulate desired endogenous gene expression dynamics, levels and regulatory inputs. Targeted transgene integration into endogenous loci offers a strategy to overcome these limitations and access previously inaccessible expression patterns.

To resolve this tool gap, we developed a quasi-enhancer trap approach harnessing phiC31 integrase-mediated transgenesis into genome-integrated attP landing sites. Using CRISPR-Cas9, we insert attP sequences into the 5' UTRs of genes with complex endogenous patterns to enable targeted transgene integration of reporters and effectors, generating transgenic lines that recapitulate native expression patterns.

Our attP knockin in hematopoietic stem cell regulator runx1 (runx1{attP}) recapitulates endogenous runx1 expression and revealed early runx1-positive mesodermal progenitors as broadly hematopoiesis-primed before activating in hematopoietic stem cells. To apply quasi-enhancer traps to augment developmental signaling studies, we targeted an attp site into the 5' UTRs of the canonical Wnt signaling target gene axin2: the resulting axin2{attP} lines faithfully reproduce dynamic canonical Wnt activity and revealed new aspects of canonical Wnt signaling during early cardiac development.

Together, our work establishes attP knockins to capture endogenous gene expression patterns for generating previously inaccessible genetic tools in zebrafish.

## *Investigating the impact of ethanol on differential gene expression and alternative splicing during zebrafish development using newly assembled T-T genome*

J.A. Lahamer, J. Chariker, and C.B. Lovely

University of Louisville School of Medicine

Fetal Alcohol Spectrum Disorders (FASD) define a highly variable set of ethanol-induced developmental defects. Through animal studies, we and others have shown that genetic risk factors underlie FASD etiology and drive the cellular and molecular mechanisms of FASD. However, these studies have focused on single-gene mutations and fail to fully explain variation within and between ethanol-treated embryos. Next-generation sequencing paired with liquid chromatography-mass spectrometry (LC-MS) provides an opportunity to identify ethanol-induced alterations in gene expression and transcript modifications leading to differential protein expression and structure. Here, we analyzed bulk RNA sequencing and long-read isoform sequencing data using a newly assembled, telomere-to-telomere annotated zebrafish genome to identify differentially expressed genes and alternative splicing in untreated and ethanol-treated zebrafish embryos at 24 and 36 hours post-fertilization. The transcriptomic data were integrated with LC-MS to identify protein-coding isoforms and validate protein-coding alterations. We found that developmental time was the main contributor to gene expression changes and epigenetic modifications. Ethanol contributed to variation in both gene expression and alternative splicing to a lesser degree. However, impacts on gene expression, splicing, and isoform usage were largely distinct from changes due to developmental time, suggesting that ethanol elicits unique responses during development. Furthermore, we observed no overlap between varying isoform usage and differentially expressed genes, raising questions of ethanol's exact role in epigenetic regulation. Together, our transcriptomic and proteomic analyses provide new insights into the complexities of gene regulation and function in FASD, and thus a foundation for future functional studies.

## *Agentic artificial intelligence for bioinformatics automation*

Yuan Lu

DeConvo Genetics

Biology has become a data-driven science. The complete curation of biological datasets, especially high-throughput sequencing data, requires multiple tools chained together to form a workflow. Manual orchestration is unsustainable, therefore workflow managers such as Nextflow, Snakemake, user interface

based Galaxy, and many custom managing pipelines have been developed. However, the main obstacle to the broad adoption of such tools by experimental biologists remains to be a lack of computational training. To facilitate high throughput data analyses, we developed Daedal. This artificial intelligence (AI)-empowered bioinformatics agent comprehends user natural language inputs, builds custom job flows with appropriate tools, executes defined jobs, handles potential failures with automatic recovery, and eventually returns fully curated results alongside a performance report. We benchmarked the agentic AI tool against workflow manager and large language model (LLM) based co-scientists to assess its tool selection accuracy and job flow reproducibility. It is shown that Daedal is capable of conducting advanced bioinformatics analyses.

### *Ethanol extract of Sutherlandia inhibits Hedgehog signaling*

Ava Garcia, Trinity Stevens, Mikki Boswell, Will Boswell, Rachel Carroll, William Folk, Zhao Lai, Wes Warren, Yuan Lu

Texas State University

Hedgehog (Hh) signaling pathway is associated with multiple types of cancers, such as melanoma and basal cell carcinoma. The activation of Hh signaling leads to remodeling of tumor niche and create a microenvironment facilitating metastasis. Sutherlandia frutescens (Sutherlandia), known as cancer bush, is a traditional South African shrub used as an herbal tonic to support immune function. Ethanol extract of Sutherlandia showed anti-hedgehog signaling activity using a reporter assay measuring Hh signaling downstream transcription factor Gli1. To study if Sutherlandia can suppress Hh in melanoma and cancer, we treated hybrid Xiphophorus fish developing melanoma by swimming the fish in aquarium water supplemented with vehicle control, Sutherlandia ethanol extract, and Hh signaling inhibitor Vismodegib, and assessed Hh signaling pathway activity using quantitative real time PCR, bulk and single nucleus transcriptome profiling. It is shown that Hh signaling are mostly present in fibroblast cells within the melanoma microenvironment, and Hh signaling related genes, including the canonical genes reflecting pathway activity gli1 and ptch1, are suppressed by Sutherlandia extract or Vismodegib compared to vehicle control. These observations suggest Sutherlandia may be beneficial for melanoma treatment through inhibition of Hh signaling.

## *Aquatic germplasm and genetic resources centers (AGGRC): the future of germplasm repositories in intersectoral and interdisciplinary*

Irene Marti Gil, Jack C. Koch, Yue Liu, and Terrence R. Tiersch
LSU AgCenter

The conservation, storage, and distribution of genetic resources from valuable livestock species constitute multiple multibillion-dollar global industries. Although fish and shellfish are vital to global economies, bioscience development, and human livelihoods, comparable systems for protecting and commercializing germplasm of aquatic species are severely underdeveloped. To bridge the gap, the Aquatic Germplasm and Genetic Resources Center was founded in 2014 within the LSU Ag Center to provide global leadership in protection, management, and commercialization of aquatic genetic resources through germplasm repository development. The ultimate goal is to support aquaculture, biomedical research, natural fisheries, conservation of imperiled species, and food security through intersectoral and interdisciplinary partnerships. As scientific and industrial communities increasingly recognize the importance of germplasm repositories and the urgency to develop them, the AGGRC has broadened its scale and scope into a conglomerate comprising three new Centers, all closely intertwined yet highly specialized. The Aquatic Germplasm Center of Research Excellence serves as a resource hub to support industry and community needs through technology development and expanded access for protecting aquatic genetic resources. The National Aquatic Germplasm Center provides services and products relevant to repository development, such as sample processing training and cryopreservation kits. The Global Aquatic Legacy Alliance bonds together a global community dedicated to protecting aquatic genetic resources through education, outreach, philanthropy, and collaboration. The combined capabilities, expertise, and focus of these four centers within a unified mission create a unique and comprehensive approach to developing germplasm repositories worldwide and serving the diverse needs of the communities they support.

## *Developing genetic and behavioral tools to study microglial biology using zebrafish*

Brandon Martz, Emilio Méndez Scolari, Jesus Portillo, Tanushree Chakraborty, Harini Iyer

Rice University

Neurological disorders are frequently accompanied by neuroinflammation and memory impairment. Here we exploit the experimental advantages of the vertebrate model organism zebrafish to gain insight into brain disorders through two complementary directions. Our first aim focuses on the resident immune cells of the brain, or microglia, which regulate brain development and neuroinflammation. Microglia contribute to numerous brain disorders, yet their cell biology remains poorly understood. In a CRISPR mutagenic screen that will be followed by mechanistic experiments in stable mutants, we identify novel regulators of microglial biology. In our second aim, we develop a learning and memory assay in zebrafish, which improves existing methods by reducing time and cost. Using this assay, we show that juvenile zebrafish learn and remember aversive stimuli and that this behavior is abolished by inhibition of glutamatergic signaling. Collectively, our research will advance the understanding of microglial cell biology and establish a new platform for investigating learning and memory in brain disorders using zebrafish.

## *Xenopus-based seizure modeling of neurological disorders*

Bridget Maynard, Erica Gardner, Varenya Sethumadhavan, Conor Smith, Rahul Nihalani, Roman Flores, Linda Quach, Eleni Pitsiniaga, Mikayla Reitsma, Viola Fanfani, Frederic Vigneault, Richard Novak

Unravel Biosciences

Unravel models neurodevelopmental disease states in xenopus laevis by leveraging its SquishyWare™ in vivo drug screening platform. A high-fidelity Cas9 is used to knockout target genes across a broad range of disorders. Clinically relevant metrics are assessed using hundreds of tadpoles, reducing the discovery to validation loop to approximately 3 months. The disease model's phenotypes are assessed across metrics including development, vestibular function, motor function, and seizure activity. Seizures are a prominent symptom across many disorders evaluated in this model including MECP2, SPATA5, NEXMIF, CSNK2A1 among others. Seizure-associated and abnormal movement assays are performed in a confined environment (24-well plate). Abnormal movements are quantified as deviations in the magnitude of motion magnitude. Wildtype tadpoles typically exhibit minimal movement in such confined environments, such that a deviation to this

motion is easily seen. The magnitude in motion is quantified using optical flow analysis of a 20 minute video recording session, and output as a continuous measure of movement intensity over time. In addition to measuring spontaneous erratic motions, the assay is performed in the presence of 0.5 mM 4-Aminopyridine, a non-selective voltage-gated potassium channel blocker and pentylenetetrazol, a GABA-A receptor antagonist inducing tonic-clonic seizures defined by erratic darting and circling movements. This assay replicates the seizure patients experience and enables the screening of potentially therapeutic compounds.

### *The role of dmrt1 in sex determination and testis development in zebrafish*

Biby Medrano, Raymond Poirier, Jocelyn Steinfeld, Keith Ameyaw, Andrew Cogliano, Jessica Periello, Andrew Karam, Joaquin Walker-Roman, Christopher Wood, Kellee Siegfried

University of Massachusetts Boston

Expression of specific genes influences sex determination and gonad development across species. For example, double sex and mab3 related transcription factor 1 (dmrt1) is a key regulator of male sex determination and testis development in many animals. In zebrafish, dmrt1 is expressed in the Sertoli cells and germ cells within the testis. Dmrt1 mutant zebrafish populations are strongly female biased and the males that do develop are sterile with aberrant testis morphology, indicating dmrt1's role in male development. However, its function in the Sertoli cells and germ cells is not entirely understood. To test if dmrt1 is sufficient to drive male sex determination and investigate its function in specific cell types, we created dmrt1 transgenes expressed in either the Sertoli cells or germ cells. These transgenes allow us to properly isolate dmrt1 expression to either the germ cells or Sertoli cells. Immunofluorescent antibody labeling confirms the proper expression of the Sertoli and germ cell transgenes. Using these transgenes, we determined that dmrt1 overexpression specifically in the germ cells was sufficient to drive male sex determination. Next, we aim to uncover which cell types require dmrt1 expression for male sex determination by analyzing dmrt1 loss of function mutants that carry a copy of either transgene. Furthermore, histology will be performed on transgenics to uncover which cell types require dmrt1 for testis development. This study will further elucidate the function of dmrt1 in sex determination and gonad development.

### *Leveraging on evolutionary adaptation to melanoma for understanding cancer resilience*

Jessica Parker, Will Boswell, Rachel Carroll, Maggs X, Wesley C. Warren, Manfred Schartl, Yuan Lu

Texas State University

Melanoma remains one of the deadliest forms of cancer due largely to high rates of intrinsic and acquired drug resistance. Despite recent therapeutic advances including immune checkpoint inhibitors and small-molecule kinase inhibitors, patients frequently and quickly develop resistance to clinically available treatments. Novel models are needed to devise strategies that overcome resistance and further reduce melanoma-related mortality. Interspecies hybrid fish from the Xiphophorus lineage develop mutant Epidermal Growth Factor Receptor (EGFR)-driven melanomas that display morphology, bulk gene expression, and recently studied cancer microenvironment mimicking those of human melanomas. It has been shown that some Xiphophorus species are resistant to melanoma despite expressing the oncogene driving disease in hybrids. We have previously uncovered genetic loci (e.g., rab3d, adgre5, and long neurotoxin OH-57-like) which are relevant to this melanoma resistance. However, the mechanism underlying the disease resilience remains unclear. To address this, we performed single-nucleus RNA sequencing (snRNAseq) analysis of Xiphophorus melanomas (MM) and healthy cancer-resistant dorsal fin (CRDF), where the melanoma initiates. Melanoma microenvironment change is characterized by a loss of mucosal cells and an enrichment of cancer associated fibroblasts, macrophages, T cell subpopulations, and B cells in MM. Cancer resistance related genes have not been found expressed in melanocyte or melanoma cells in CRDF or MM, suggesting the cancer resilience is mediated through inter-cellular interactions as opposed to cell autonomous processes.

### *The mystery of zebrafish gonad development: a graphic novel*

John H. Postlethwait and Kristopher Chu

University of Oregon

Public understanding of scientific research is an increasingly important problem when dark forces conspire to limit exploration of how the natural world works. Graphic novels can help bring science and how-we-know-what-we-know to audiences that often find science to be inaccessible. To help broaden access, we have begun to develop comics that explain our work. In this graphic novel, led by an undergraduate majoring in Art and Technology, we pose the question: How do zebrafish gonads decide to become an ovary or

a testis? Touching on the interplay between environment and genetics, the comic raises questions, presents hypotheses, describes experiments, illustrates results, draws conclusions, explains significance, and suggests further work. The comic is suitable for high school classrooms and more. It is freely available to the public for download at (https://zenodo.org/records/16790438). Our earlier comic entitled “What controls sex development in fish” is also available for download in nine languages at (https://zenodo.org/records/13737141).

## *Nesprin-1 is required for establishment of left-right asymmetry in zebrafish*

Aaliyah M. Ruiz-Corral , Emma H. Gerlt, Po-Shu Tu, Stefan C. Materna, Ph.D.
University of California, Merced

The zebrafish left-right organizer, Kupffer’s Vesicle (KV), is a fluid-filled vesicle composed of a simple epithelium in which motile cilia generate directional flow that ultimately establishes asymmetric gene expression. Single-cell RNA sequencing identified syne1b, which encodes the giant KASH-domain protein Nesprin-1, as the most highly enriched gene in KV and its progenitors, the dorsal forerunner cells (DFCs). Nesprin-1 is a component of the LINC (Linker of Nucleoskeleton and Cytoskeleton) complex, in which its C-terminal KASH domain anchors the protein to the outer nuclear membrane, where it interacts with SUN proteins in the inner nuclear membrane to couple the cytoskeleton to the nucleus. Although Nesprin proteins have been implicated in processes such as mechanosensation, cellular scaffolding, signaling, and nuclear positioning, their role during KV morphogenesis remains poorly understood. Gene-editing studies demonstrated that Nesprin-1 is required for proper KV development, as loss of Nesprin-1 disrupts KV morphogenesis, impairs lumen formation, alters nuclear morphology, and ultimately perturbs left-right patterning. We hypothesize that Nesprin-1 functions through the LINC complex to transmit mechanical forces to the nucleus, thereby coordinating nuclear organization and downstream cellular behaviors required for proper KV morphogenesis and left-right patterning.

## *Introduction of a new scale to analyze ethanol sensitivity of compound Bmp and Fgf double mutants*

Hieu Vo, Ben Lovely

University of Louisville

Fetal alcohol spectrum disorders (FASD) describe a wide range of neurological behaviors and physical features of an individual exposed to alcohol prenatally. While we have a greater understanding of the genetic loci underlying FASD, the interplay between these loci is poorly understood. Here, I examined an ethanol-sensitive molecular pathway that constitutes facial development. We have previously shown that a Bmp-Fgf signaling pathway drives facial development through endoderm morphogenesis. This Bmp-Fgf-dependent pharyngeal endoderm morphogenesis occurs between 10 and 36 hours post-fertilization (hpf). We have also shown that Bmp regulates Fgf signaling by modulating fgfr4 expression, and blocking Bmp signaling from 10-18 hpf reduces endodermal Fgf signaling, leading to defects in endoderm morphogenesis and facial development. However, the exact interplay between components of each pathway and ethanol's impact is still unclear. Here, using Bmp-Fgf compound mutant zebrafish, I examine the facial skeleton of the untreated and ethanol-treated (10-18 hpf) larvae at 5 days post-fertilization (dpf). I show that different mutations in Bmp and Fgf signaling pathway components differentially sensitize embryos to ethanol. While bmp4 and fgf8a highly sensitize embryos to ethanol, smad5 and fgf3 tend to have a fixed facial phenotype, independent of ethanol, and are usually the deciding genes for the facial phenotype severity. I organize and present a "severity scale" of each gene based on the untreated and ethanol-treated facial defects. Overall, my work shows the multifaceted interplay between complex signaling pathway components and ethanol and provides insight into ethanol's impact on signaling dynamics.

## *Dnmbp interacts with Daam1 to facilitate assembly of cadherin-mediated junctions in epithelializing nephric tubules*

Brandy L. Walker, Bridget D. De Lay, Yogesh Srivastava, Mark E. Corkins, Vanja Krneta-Stankic, Adrian Romero, and Rachel K. Miller *

UTHealth Houston

The mature kidney contains approximately one million nephrons, and defects arising during nephron development can result in lifelong renal impairment, often culminating in kidney failure and transplantation. Nephric tubule formation requires coordinated epithelial processes, including polarity, adhesion, signaling, and vesicle transport; however, how these processes are

integrated during kidney morphogenesis remains unclear. Dynamin binding protein (Dnmbp) is a multi-domain scaffolding protein expressed in human kidneys that is involved in several cellular processes. Using the Xenopus embryonic kidney, we previously demonstrated that Dnmbp is essential for nephrogenesis, yet the mechanisms by which it influences nephron development remain undefined. Here, we identify Dnmbp as a novel interacting partner of the Wnt/planar cell polarity effector Daam1. The interaction between Daam1 and Dnmbp was independently identified in two yeast two-hybrid screens, biochemically verified, and supported by structural modeling predictions of a Daam1-Dnmbp complex. In developing Xenopus laevis kidneys, Dnmbp localized to punctate structures associated with E-cadherin-rich cell-cell contacts. Dnmbp depletion significantly reduced junctional E-cadherin localization in both epithelializing and mature nephric tubules without affecting total E-cadherin levels, indicating a role in E-cadherin recruitment or stabilization at adherens junctions. Furthermore, expression of human DNMBP rescued the junctional defects, confirming the specificity of the loss-of-function phenotype. Together, these findings identify Dnmbp as an essential regulator of kidney development and support a model in which Dnmbp provides a mechanistic link between Wnt/PCP signaling, Cdc42 activation, and adherens junction formation during nephrogenesis.

## *Towards rapid generation and ready access of monoclonal antibody reagents*

Linjian Wang, Chris Amemiya

University of California, Merced

Monoclonal antibodies (mAbs) are invaluable reagents for our community and are used extensively in basic biological research and in the biomedical industry. mAbs comprise a multi-billion-dollar industry annually, but the generation of these reagents is a very expensive and labor-intensive proposition for most academic researchers. An expedient and inexpensive platform to produce robust and stable monoclonal antibodies that can be distributed using an open-source model is highly desirable. To this end, we submit that we can exploit the lamprey immune system, innovative molecular cloning methods, and synthetic biology to rapidly and inexpensively generate robust lamprey-based antibodies (lampribodies) to protein and glycan targets.

Lampreys have evolved a convergent adaptive immune system compared to that of the jawed vertebrates (e.g., mouse, human) that employs leucine-rich repeats rather than VDJ segments for building their antibody repertoire. It is estimated that a staggering 10^14 unique lamprey antibodies are theoretically

possible, similar to that of the mouse antibody repertoire. We are developing a novel system that expresses lampribodies on the inner membrane of E. coli cells that should greatly facilitate screening and purification for specific antigen binders. Purified lampribodies will then be conjugated to oligos for use in immunohistochemistry-hybridization chain reaction (IHC-HCR). This platform would be much more expedient than the current methods for generating monoclonal antibodies and would provide a useful and affordable tool for various areas of research and medicine.

## *Exposing dynamic functions of KCNH1 underlying disease etiologies and morphological variation*

Miranda L. Wilson, João Castro, Cure KCNH1 Foundation, Matthew P. Harris
Boston Children's Hospital

Ion channels serve an essential function by regulating a cell's ionic concentration, and consequently its membrane potential, to support a variety of functions and physiological homeostasis. It is therefore unsurprising that altered ion channel function would have severe outcomes on both a cellular and organismal level. We have uncovered unique developmental functions regulated by classes of potassium channels that fall outside of their canonical activities. Gain-of-function (GOF) variants of the voltage-gated potassium channel Kv10.1, encoded by KCNH1/EAG1, have been associated with KCNH1-related developmental disorders. Affected individuals exhibit a range of phenotypes, including early-onset epilepsy, intellectual dysfunction, global developmental delay, gingival overgrowth, and hypoplasia of the phalanges. In large experimental genetic screens in the zebrafish, we have identified GOF mutations in the KCNH1 ortholog, kcnh1a, that regulate the patterning and coordinated growth of fins. Importantly, these fish do not display neurological abnormalities such as seizures. Zebrafish contain an additional KCNH1 ortholog, kcnh1b, though GOF mutations have yet to be reported and thus the impact of such variants in the brain and fins remains unknown. Using a targeted gene editing approach, we will generate fish harboring patient mutations in kcnh1a and kcnh1b to further characterize the mechanisms that cause these pathologies. Further, mutant fish exhibiting seizures will be utilized in a small molecule drug screen to isolate candidates for the treatment of neurological symptoms associated with GOF KCNH1 mutations. Altogether, this work will better define Kv10.1 potassium channel dysfunction in the brain and skeletal system and possibly define treatments for affected individuals.

# Poster Session II

## *Developmental impacts of harmful algal bloom (HAB) toxins on the vertebrate nervous system*

Neelakanteswar Aluru, Racliffe Lai, Mekia Bushell, Mark E. Hahn
Woods Hole Oceanographic Institution

HABs are increasing worldwide, leading to greater human exposure to neurotoxic cyanobacterial and marine algal toxins. Among these, saxitoxin (STX), a voltage-gated sodium channel blocker, and (+)-anatoxin-a (ATX-a), a nicotinic acetylcholine receptor agonist, disrupt neuronal signaling through distinct modes of action, yet their effects on vertebrate neurodevelopment remain poorly understood. We use zebrafish to investigate the cellular and molecular mechanisms underlying developmental neurotoxicity following early-life exposure to these toxins. Embryos were exposed to STX or ATX-a by microinjection during defined stages of embryogenesis, followed by assessments of morphology, behavior, neurodevelopment, and gene expression. Developmental STX exposure produced subtle developmental and behavioral abnormalities and altered molecular pathways involved in neuronal differentiation, ion channel function, and extracellular matrix organization. To resolve cell type-specific responses, we are focusing on sensory and motor neurons and performing single-cell RNA sequencing of the brain to identify cellular targets of STX. In contrast, developmental ATX-a exposure produced rapid but reversible paralysis, impaired touch responsiveness, transient motor axon abnormalities, and robust transcriptomic changes enriched for skeletal muscle development, neuromuscular signaling, and early neurodevelopment. These findings demonstrate that the distinct molecular targets of STX and ATX-a translate into different cellular, transcriptional, and functional consequences during vertebrate development. By integrating behavioral, cellular, and transcriptomic approaches, this work provides mechanistic insight into HAB toxin-induced developmental neurotoxicity. Given the increasing potential for human exposure to HAB toxins, defining these conserved developmental responses in zebrafish may help identify mechanisms and cellular targets relevant to human neurological health following early-life exposure. [P01ES208938, OCE-2418297]

## *Investigating the neuroendocrine axis of prolactin in the development of Xenopus*

Eva Beiga, M. Clayton Speed, Jessica Benitez, Valérie Langlois, Marko Horb
Marine Biological Laboratory

African clawed frogs; Xenopus are a valuable vertebrate model for developmental and cellular biology. At the Xenopus Mutant Resource (XMR), we utilize Xenopus tropicalis and X. laevis to generate mutant animals with specific gene knockouts. Work is done in collaboration with researchers that are interested in using Xenopus as a model for aspects of human disease and development. Prolactin is an essential hormone for human development, with alteration in its abundance causing developmental deficits, such as loss of lactation and hormonal imbalance. Using the model vertebrate X. tropicalis, we seek to investigate the neuroendocrine role the prolactin pathway plays in developmental timing. It has previously been reported that prolactin supports larval growth by acting as a functional antagonist to thyroid hormone, which drives metamorphosis and tissue differentiation. If the prolactin gene is knocked out, we predict developmental acceleration and faster sexual maturity due to the lack of prolactin's antagonistic effects. Using CRISPR/Cas9 mutagenesis, we have generated mutant animals which are defective in their production of prolactin. In our preliminary studies, we see evidence of altered developmental timing. To validate these results, we are now generating animals which lack the prolactin receptor; to reciprocate the phenotype of prolactin deficiency. Taken together, these experiments will establish an animal model to investigate the Prolactin-Thyroid pathway and its role in developmental timing and disease. This is only one example of the types of projects we perform in collaboration with other researchers. The XMR generates thousands of mutants to serve the larger scientific community.

## *Developing killifish models for functional and comparative genetics*

Petra Bora, Andrew R. Kaczmar, Andrew W. Thompson
Western Michigan University

While traditional laboratory models such as zebrafish, mice, and medaka have advanced vertebrate genetics, comparative approaches in functional genetics remain limited across vertebrate diversity. However, some vertebrates, such as killifishes, particularly “annual” species that inhabit seasonal pools and live for about a year or less, offer valuable models because of their short life cycles and developmental dormancy, making them ideal for studying genetic, developmental, and age-related variation. Although the rapidly aging African

turquoise killifish has emerged as a model for aging and developmental biology, most of the more than 800 other killifish species remain largely unexplored. Thus, we are developing the green rivulus, Rivulus cylindraceus, and Rio pearlfish, Nematolebias whitei, as genetically tractable laboratory models for evolutionary and functional genetics. As a proof of concept, we used CRISPR-Cas9-mediated targeting of the pigmentation gene oculocutaneous albinism 2 (oca2) in R. cylindraceus, generating partial oca2 knockout embryos. These embryos exhibited reduced pigmentation, demonstrating successful gene disruption. We also generated single-nucleus RNA-seq datasets in N. whitei to characterize cell-type-specific gene expression across biological contexts, including sex-specific expression programs in the liver. Together, these findings validate genome editing in R. cylindraceus and establish a foundation for applying functional genomic and transcriptomic approaches across these and related killifish species. Thus, the non-annual R. cylindraceus and annual N. whitei represent valuable, complementary models for comparative functional genetics and genomics.

## *Xenopus models to study genetic function in human pathology*

Kelsey Coppenrath, Nikko-Ideen Shaidani, Thomas Naert, Himani D. Majumdar, Soeren S. Lienkamp, Steven L. Klein, Federico Gonzalez-Fernandez, Sally A. Moody, Artur Llobet, and Marko Horb

Marine Biological Laboratory

The Xenopus Mutant Resource (XMR) provides an extensive genetic library for the scientific community to conduct tractable experiments on gene function and the development of human diseases. One of the goals of the XMR is to work collaboratively with scientists by generating Xenopus models through targeted CRISPR injections to help facilitate their work. A recent collaboration demonstrated that one such gene, lamp2 in X. tropicalis, displays histological and functional characteristics of Danon Disease, an X-linked genetic disorder resulting in cardiomyopathy, skeletal myopathy and intellectual disability. In X. tropicalis, a previously characterized mutant line six1, was utilized in a recent study to demonstrate the effects two Branchio-oto-renal (BOR) syndrome variants have on early development and morphogenesis. Variants arising from mutations in either the protein-protein interacting domain (SD) or homeodomain (HD) were expressed in six1 reduced backgrounds to compare gene expression in tissues affected by BOR; such as the neural crest, preplacodal ectoderm, and otic vesicles. Additionally, we are currently developing knock-out mutant RBP3 in both X. tropicalis and laevis that is associated with retinitis pigmentosa, a group of eye disorders which lead to deterioration of photoreceptors in the retina. Establishing an RBP3 knockout

line in Xenopus will aid in understanding its role in early ocular development, as previous studies have primarily focused on its function at adult life stages. Here we present our work in generating knockout mutant lines at the XMR to study disease phenotypes in various tissues. Work done here is supported by the NIH R24 OD030008.

## *Spatial characterization of egg-laying hormone and caudodorsal cell peptides in the central nervous system of Biomphalaria glabrata*

Jonathan L. Crooke-Rosado, Natali M. Colón, Mark W. Miller
University of Puerto Rico

Background: Reproductive regulation may be a critical point of interaction between Biomphalaria glabrata and its trematode parasite Schistosoma mansoni, the causative agent of intestinal schistosomiasis. Egg-Laying Hormone (ELH) and the related Caudodorsal Cell Hormone and Peptides (CDCH/CDCP) are neuropeptides associated with the neural control of reproductive physiology in gastropods. Because S. mansoni infection can profoundly alter host reproductive investment, defining the neural systems controlling egg-laying is essential for understanding parasite-induced reproductive changes.

Knowledge Gap and Rationale: The cellular distribution and expression patterns of ELH and CDCH/CDCP in the B. glabrata central nervous system (CNS) remain poorly characterized. Establishing their baseline organization at both transcript and protein levels provides a foundation for determining how these reproductive pathways change following infection.

Methods: In situ Hybridization Chain Reaction (HCR), immunofluorescence, and confocal microscopy were combined to characterize neuropeptide mRNA and protein localization throughout the CNS.

Results: ELH transcripts are predominantly localized to the cerebral ganglia, including an intensely labeled bilateral dorsolateral cell cluster and two to four less-intense ventral clusters showing left-right asymmetry. ELH/CDCH-related immunoreactivity was prominent in neuronal fibers forming a ring-like network between the dorsal ganglia and in neuronal somata corresponding to regions of strong mRNA expression.

Conclusion: These findings provide a neuroanatomical framework for investigating infection-associated changes in reproductive neuroendocrine signaling in B. glabrata and their potential contribution to host–parasite interactions.

## *On the evolution of the aryl hydrocarbon receptor (AHR), a promiscuous sensor of xenobiotic and endobiotic chemicals*

Mark E. Hahn, Jared V. Goldstone, Sibel I. Karchner, Tessa Steiniche, Joseph W. Thornton

Woods Hole Oceanographic Institution

The aryl hydrocarbon receptor (AHR) is a pleiotropic bHLH-PAS transcription factor that mediates responses to toxic environmental contaminants such as chlorinated dioxins, but the AHR also regulates physiological processes including development, immunity, and host-microbiome interactions. The vertebrate AHR is highly promiscuous, recognizing diverse chemicals, including natural products such as dietary phytochemicals, microbiome-derived metabolites, and tryptophan catabolites. To better understand the physiological and toxicological roles of AHR, we are investigating its molecular evolution and reconstructing the evolution of its ligand-dependence and promiscuity. We identified 677 full-length AHR amino acid sequences, including AHR orthologs in pre-bilaterian phyla (placozoans; cnidarians), ten protostome phyla, non-chordate deuterostomes (echinoderms; hemichordates), invertebrate chordates (tunicates; cephalochordates), jawless fishes (lamprey, hagfish), cartilaginous fishes, bony fishes (cladistians, chondrosteans, holosteans, teleosts), and tetrapods (amphibians, reptiles, birds, mammals). Phylogenetic analyses demonstrated that most invertebrates have a single AHR, whereas vertebrate animals possess multiple AHRs that fall into six major clades (paralog groups): AHR0, AHR1, AHR2, AHR3, AHR4, and AHRR. Our results show that AHR is an ancient protein that is present in nearly all modern metazoans, including some of the earliest diverging taxa. AHR genes have undergone duplication and diversification in the vertebrate lineage, coinciding with the diversification of their ligand-binding properties, including the emergence of high-affinity binding of dioxins and other xenobiotics. Understanding the evolutionary history of the AHR and its ligand-binding properties will provide new insights into the natural physiological ligands and biological functions of this protein and the roles of AHR in human disease. [Supported by NIH R01ES035261.]

## *Utilizing transgenic zebrafish to screen per- and polyfluoroalkyl substances (PFAS) for potential immunotoxicity*

Emma Marie Wilber Hepworth and Jeffrey A. Yoder

North Carolina State University

Per- and polyfluoroalkyl substances (PFAS) are a class of chemicals that have become widespread and persistent pollutants due to their extensive use in numerous consumer products and unique chemistry. There is substantial evidence that exposure to PFAS is linked with multiple adverse health outcomes, including altered immune function. Due to the vast number of PFAS that lack sufficient immunotoxicity data, there is significant need for new high-throughput screens that can rapidly assess how PFAS exposure impacts the immune system. Zebrafish (Danio rerio) are an established experimental model used in translational toxicology and have proved to be a versatile tool for studying the effects of PFAS exposure. Two immunology-focused transgenic zebrafish lines were utilized for these experiments: Tg(NFkB:EGFP), which expresses a general marker for immune stimulation, and Tg(lyz:EGFP), which expresses fluorescently labeled neutrophils. Larvae were exposed to different PFAS for a total of 5 days and were imaged at 3, 4, and 5 days post fertilization to assess larvae morphology and measure either fluorescence intensity in the Tg(NFkB:EGFP) larvae or neutrophil number in the Tg(lyz:EGFP) larvae. Preliminary results indicate that multiple PFAS tested induce at least some degree of immunotoxic effects in larval zebrafish. However, the exact pattern of these effects differs amongst the PFAS. Since PFAS are an exceptionally large and structurally diverse class of chemicals, individual PFAS can have potentially varying effects and toxicity levels. This further highlights the importance of high throughput screens such as this for assessing PFAS immunotoxicity.

## *A single-nucleus transcriptomic atlas of post-organogenesis diapause in a vertebrate*

Andrew R. Kaczmar, Petra Bora, Andrew W. Thompson

Western Michigan University

Diapause is a reversible arrest of embryonic development characterized by reduced metabolism, restricted growth, and enhanced stress tolerance. This adaptation is widespread across animal lineages and allows organisms to tolerate adverse environmental conditions until development can resume. Annual killifishes have evolved to survive seasonal desiccation of their aquatic habitats through three distinct diapause stages that occur at different points in development. Consequently, annual killifishes have become important

vertebrate models for studying dormancy, aging, and developmental plasticity. Previous research has largely focused on aging and on diapause II, which occurs before organogenesis is complete. However, little is known about post-organogenesis diapause (diapause III), its effects on fully formed organ systems, and the transition from diapause to a metabolically active, hatched state. To address this gap, we developed a whole-organism single-nucleus transcriptome atlas comparing diapause III embryos and hatched larvae of the South American annual killifish Nematolebias whitei. N. whitei is an ideal model for studying post-organogenesis diapause because it undergoes a prolonged and deep diapause III that is terminated by inundation. Furthermore, N. whitei represents an independent evolutionary origin of diapause relative to other killifish models such as Nothobranchius and Kryptolebias. Overall, this atlas provides an important resource for characterizing cell types, identifying cell-type-specific expression patterns, and investigating the molecular basis of post-organogenesis diapause and hatching.

## *Studies on the role of AIP (AHR-interacting protein), a human tumor suppressor, in controlling sensitivity and resistance to aryl hydrocarbon receptor (AHR) agonists*

Sibel I. Karchner, Bryan W. Clark, Diane Nacci, Noah M. Reid, Andrew Whitehead, Mark E. Hahn

Woods Hole Oceanographic Institution

Humans vary in their sensitivity to chemicals such as 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) and polychlorinated biphenyls (PCBs) that cause toxicity through the aryl hydrocarbon receptor (AHR). A clue to the mechanisms underlying differential sensitivity to such dioxin-like compounds (DLCs) has emerged from studies of Atlantic killifish (Fundulus heteroclitus) populations that have evolved resistance to DLCs. Genomic studies identified AHR-interacting protein (AIP) as the strongest candidate resistance gene, but its role in the mechanism of resistance is unknown. AIP is a susceptibility gene for human Familial Isolated Pituitary Adenoma (FIPA); dozens of single-nucleotide variants (SNVs) in AIP predispose to this disease. We identified AIP SNVs in PCB-sensitive and PCB-resistant populations of killifish. To study the function of these variants, we transiently transfected them into COS-7 cells along with AHR and a luciferase reporter gene and then measured TCDD-inducible luciferase activity and nuclear localization of YFP-tagged AHR. Wild-type killifish or human AIP retained AHR in the cytoplasm and reduced both basal and TCDD-inducible luciferase activity. Human AIP variants identified in FIPA patients or shown previously to exhibit reduced binding to AHR did not

affect transactivation. In contrast, killifish AIP W224L and its human counterpart AIP W223L displayed enhanced function, retaining AHR in the cytoplasm and strongly reducing both basal and TCDD-inducible luciferase expression. The results suggest that variation in AIP protein structure can affect the sensitivity of the AHR signaling pathway. Understanding how AIP controls sensitivity to AHR agonists may shed light on the role of AIP in human disease. [P42ES007381, R01ES032323, R01ES033888]

## *The agrochemical forchlorfenuron produces early developmental defects in zebrafish embryos*

Gurleen Kaur and Stephanie Woo

University of California, Merced

Repetitive and non-judicious use of agrochemicals can result in pollution, a growing area of concern for wildlife and human health. One such agrochemical is forchlorfenuron (FCF), also known as CPPU (N-(2-chloro-4-pyridinyl)-N'-phenylurea), widely used to enhance the size and weight of fruits such as grapes, kiwifruits, and melons. Despite reports suggesting FCF can bioaccumulate to hazardous levels, in vivo mechanistic details of FCF toxicity remain unknown. Here, we report that treating zebrafish (Danio rerio) embryos with FCF produces early developmental defects. Embryos exposed to 50 µM and 75 µM FCF exhibited defects in epiboly, developed into a pronounced ovoid shape, and were significantly shorter and wider compared to control embryos. These phenotypic defects bear striking similarity to zebrafish mutants with impaired Bone Morphogenic Protein (BMP) signaling. Using a combination of immunostaining for phosphorylated Smad5 and qPCR expression analysis of downstream BMP target genes, we are currently determining whether FCF treatment affects BMP signaling, which has not been previously demonstrated.

## *Investigating programmed genome rearrangement across divergent lamprey life histories*

Sam Lampman, Petra Bora, Devin Bloom, Andrew W. Thompson

Western Michigan University

Lampreys are a group of jawless fishes that diverged early in the evolution of vertebrates. Jawless fishes are excellent models for developmental genetics and increase our understanding of how vertebrates, including humans, have developed and evolved. For example, jawless fishes possess neural crest cells,

a key innovation in the evolution of the vertebrate body plan. All extant lamprey species exhibit one of three life history strategies: 1) migratory and parasitic, 2) non-migratory and parasitic, or 3) non-migratory and non-parasitic. This diversity includes closely related, genetically similar species with vastly different adult life histories. Additionally, developing lampreys undergo programmed genome rearrangement (PGR), which consistently eliminates a specific subset of the genome from somatic tissues during early development. The complete genome is retained only in the germline. Here, we focus on two sister species: the chestnut (Ichthyomyzon castaneus) and southern brook (Ichthyomyzon gagei) lampreys. Neither species is migratory; however, after metamorphosis, chestnut lampreys are parasitic, feed, and possess an intact gut, while southern brook lampreys are non-parasitic, do not feed, and lack a functional adult gut. We sequenced the gametic and somatic genomes of both species and are using computational approaches to identify genetic elements that are conserved or differ between species and between germline and somatic genomes following PGR. These comparisons will provide a foundation for investigating how programmed genome rearrangement may contribute to developmental and life-history divergence between closely related lamprey species, including differences in adult feeding and digestive tract morphology.

## *Gene-ethanol interactions hint at complex pathway dynamics in facial development*

J.R. Klem, T. Schwantes-An, M. Suttie, R. Gray, H. Vo, G. Conley, L. Wetherill , CIFASD, and C.B. Lovely

University of Louisville School of Medicine

Fetal Alcohol Spectrum Disorders (FASD) describe a continuum of ethanol-induced developmental defects, including facial malformations. While ethanol-sensitive genetic mutations contribute to facial malformations, the impacted mechanisms remain unknown. Formation of the facial skeleton requires complex signaling interactions between the neural crest (NC), which forms the facial skeleton, and the endoderm, with proper endoderm morphogenesis central to these interactions. Bone Morphogenetic Protein (Bmp) signaling is a key regulator of endoderm morphogenesis. Loss of Bmp signaling results in a hypoplastic endoderm, providing a possible ethanol-sensitive mechanism. Here, we establish an in vivo FASD model, where mutations in the Bmp pathway sensitize embryos to ethanol-induced facial defects by disrupting endoderm morphogenesis. Interestingly, ethanol-treated Bmp mutants exhibit significant increases in endodermal size, similar to what we observe in knockdown of Shh signaling, a known ethanol target. Using quantitative morphometric readouts, we show that increases in endoderm

size mirror shape changes to the facial skeleton. Our zebrafish data are predictive of variants in humans with BMPR1B associating with ethanol-related differences in jaw volume. Strikingly, ethanol does not impact Bmp signaling directly but increases apoptosis in migratory NC in Bmp mutants. This suggests that Bmp-dependent signals from the endoderm are ethanol-sensitive, leading to NC defects, while Shh-ethanol interactions may drive endoderm morphology changes. Overall, we show that our zebrafish FASD model is a powerful, efficient model to simultaneously generate a deeper mechanistic understanding of gene-ethanol interactions on complex tissue interactions and provide a conceptual framework for ethanol-induced structural birth defects in humans.

## *Evolved differences in microglial cell biology between surface and cave populations of Astyanax mexicanus*

Emilio Mendez Scolari, Othniel Amanyi, Aakriti Rastogi, Erik Duboue, Alex Keene, Harini Iyer

Rice University

Microglia are brain-specific macrophages that play vital roles in brain development and homeostasis, such as phagocytosing neuronal debris, refining neural circuits, mediating immune responses, and supporting neurogenesis. Elucidating the regulatory mechanisms underlying microglial functional plasticity is critical for understanding their role under physiological and pathological conditions, and natural evolution provides a unique opportunity to investigate microglial adaptability across diverse biological contexts shaped by environmental pressures. The Mexican tetra Astyanax mexicanus has emerged as a powerful model for studying the evolution of extreme biological traits due to its existence as surface and cave morphotypes with prominent behavioral and physiological differences. The evolutionary traits acquired by cavefish, including eye loss, reduced sleep, altered brain anatomy, and diminished immune response, provide an exceptional platform to study microglial adaptation to these natural modifications. However, microglia have not been investigated in Astyanax. Therefore, we developed experimental approaches to examine microglial specification, dynamics, and function in A. mexicanus to perform a comparative analysis of microglial biology between surface fish and cavefish. We found that cave populations show increased numbers of microglia over developmental time compared with their surface counterparts. Microglia in Astyanax rapidly expand in response to inflammatory cues. Furthermore, lysosomal compartments of microglia in the cave populations exhibit enhanced proteolytic activity and reduced pH relative to surface morphotypes. Together, our observations

reveal evolved differences in microglial cell biology between surface and cave populations of A. mexicanus and provide a framework to uncover novel neuroimmune mechanisms underlying the remarkable adaptations of A. mexicanus.

## *Evolutionary dynamics and master regulator turnover in Xenopus sex determination*

Heather A. Murphy, Martin Knytl, Lindsey M. Kukoly, Sarah R. Porter, Danielle C. Jordan, Domnick Victor Wasonga, Barbora Bergelová , Nikko-Ideen Shaidani, William R. Thomas, Carl Anderson, Ian Dworkin, Ben J. Evans, and Marko Horb
Marine Biological Laboratory

Despite the fundamental requirement of sexual differentiation for sexual reproduction, the genetic triggers governing sexual differentiation evolve rapidly across many metazoans. African clawed frogs – genus; Xenopus – are a promising model system with which to study how these master switches originate and evolve. To this end, we generated knockout lines using CRISPR-Cas9 targeting several sex-related genes, including (a) the doublesex and mab-3 related transcription factor 1 (dmrt1), (b) wnt family member 3 (wnt3), and (c) androgen receptor (ar). In polyploid species such as X. laevis and X. borealis, whole-genome duplication forms paralogs that acquired distinct roles in sex-specific development. Knockout of dmrt1 homeologs in X. laevis and its diploid relative X. tropicalis reveals that dmrt1 underwent both subfunctionalization and neofunctionalization following allotetraploidization—a process that eventually produced the non-essential dmrt1.S paralog from which the female-determining dm-w gene originated. Beyond conserved pathways, lineage-specific triggers can also rapidly arise. In X. borealis, a novel, female-specific copy of wnt3 (wnt3-W) recently evolved on the tip of the W chromosome; functional knockout converts genetic females into fertile phenotypic males, marking wnt3-W as a rare example of a female-determining genetic trigger in the wnt signaling pathway. Finally, knockouts of ar in X. laevis show that targeted mutations can bypass ancestral W-chromosome control entirely, creating synthetic XY heterogametic systems capable of supporting fertile populations. Together, these findings highlight how gene duplication, neofunctionalization and subfunctionalization, and developmental systems drift allow Xenopus species to rapidly recruit novel triggers and transition between sex-determining systems without disrupting reproduction.

## *Establishing clams as an innovative model system to understand biomineralization and aging*

Indiana Olson, Justin Waletich, Stephen Treaster, Matthew Harris

Boston Children's Hospital

The oldest, non-colonial invertebrate is the ocean quahog, Arctica islandica, with some individuals documented at over 500 years old. Despite living in the same waters as other clam species like the Atlantic surf clam (Spisula solidissima), and northern quahog (Mercenaria mercenaria), ocean quahogs eclipse their lifespans by centuries. The variation in lifespan among these species provides a natural case study to address the underlying genetic encoding of extreme longevity. Through developing comparative genomic and genetic tools, we ask if similar mechanisms are conserved in other animals, including humans. We have found that clams age, one phenotype that arises is defects in mineralization of their shell. A specialized tissue, the mantle, is responsible for biomineralization of the shell; however, the precise mechanism for patterning shell components remains unknown. The surf clam, S. solidissima, represents an attractive genetic model candidate due to its readily inducible spawning of millions of gametes and short development time, around 3 months. Preliminary work has shown that electroporation can successfully load gametes with fluorescent molecules for live imaging and dsRNAs can facilitate gene knockdown. We are preparing a draft genome for this clam and have established key experimental assays for testing biomineralization. Furthermore, we capitalize on experimental accessibility of clams to directly test questions brought out by comparative genomic analyses. We hope to leverage this natural model for biomineralization, aging, and lifespan to dissect out the means by which evolution can mold life history strategies, while informing new strategies for modulating these processes in us.

## *In-vitro cultured embryos of Xiphophorus as multipurpose tools for biomedical studies*

Juliana Morena Bonita Ricci, Manfred Schartl

Texas State University

Xiphophorus fish provide several well-established and emerging models for human disease. So far only adult stages were available for research, because Xiphophorus is a member of Poeciliid fishes, which comprise a diverse group of freshwater species characterized by viviparity, in which embryos develop internally and are born as live young. This reproductive mode restricts access to early embryonic stages, creating significant barriers to genetic modification

and experimental manipulation during development. Embryo culture therefore provides a powerful approach for investigating organogenesis and enabling direct manipulation of embryos. Here, using an optimized in vitro embryo culture protocol for the genus Xiphophorus, we generated transgenic live-bearing fish via lentiviral-mediated gene delivery. Embryos were microinjected with lentiviral particles carrying a CMV-driven GFP reporter and exhibited stable transgene expression, with strong fluorescence in the brain and notochord. We also performed xenograft transplantation into developing embryos in vitro. Fish melanoma cells (PSM-1) and human skin melanoma cells (UACC257) were transplanted into commonly used injection sites (pericardial space, dorsal region, and yolk sac), where they successfully colonized host tissues locally. Moreover, the availability of Xiphophorus mutant and hybrid genotypes which develop neoplastic transformation of pigment cells already during embryonic development offers the opportunity for drug treatments of such early stages. Together, these approaches establish embryo culture as a powerful experimental platform, opening new possibilities for developmental and genetic studies in Xiphophorus.

## *Neurochemical adaptation during aestivation in the freshwater snail Biomphalaria glabrata*

Rivas Jiménez-Arelys, Rivera Rivera-Arturo, Miller-Mark

University of Puerto Rico Medical Sciences Campus

Aestivation is a reversible state of dormancy that allows organisms to survive prolonged periods of environmental challenge, particularly dehydration. The freshwater snail Biomphalaria glabrata, an established model for investigating host-parasite interactions, can survive prolonged dehydration by suppressing metabolic, physiological, and behavioral activity. While substantial research has characterized metabolic and systemic adaptations associated with aestivation, the effects of this process on the central nervous system (CNS) remain poorly understood. Here, we investigate CNS adaptations associated with aestivation in B. glabrata, focusing on changes in serotonergic signaling during and following the aestivation period. Snails were maintained under controlled environmental conditions and induced to aestivate for 14 days. The CNS and peripheral tissues were collected from active controls, aestivated snails, and animals following rehydration. Wholemount immunohistochemistry localized serotonergic neurons and projections throughout the CNS. During aestivation, serotonergic immunoreactivity was markedly reduced compared with active controls, while the major serotonergic neuron clusters remained anatomically preserved. Following reintroduction of water, snails progressively recovered baseline systemic

function and characteristic behaviors, accompanied by restoration of serotonin-like immunoreactivity within the CNS. These findings suggest that a coordinated suppression and recovery of serotonergic signaling could contribute to the transition between active and dormant physiological states. This work provides new insight into the neurobiological mechanisms underlying aestivation and recovery in an aquatic invertebrate model.

## *Transgenerational epigenetic reprogramming of germline and gonadal tissues following ancestral benzo[a]pyrene exposure in Japanese medaka (Oryzias latipes)*

Nonnie E. Cook, Leisha Martin, Remi Labeille, Frauke Seemann

Texas A&M University-Corpus Christi

Transgenerational inheritance of toxicant-induced phenotypes is an important issue in environmental toxicology and epigenetics. Benzo[a]pyrene (BaP), a ubiquitous polycyclic aromatic hydrocarbon, has been implicated in epigenetic dysregulation through altered chromatin organization and DNA methylation in germ cells. We hypothesized that BaP-induced epigenetic alterations in both paternal and maternal germlines contribute to the transmission of adverse phenotypes across generations.

To address this, sperm, testis, small oocytes, large oocytes, and ovary tissue were collected from parental and descendant Japanese medaka (Oryzias latipes) and analyzed for chromatin structure and DNA methylation. Ancestral BaP exposure significantly altered epigenetic profiles in all tissues examined. Increased heterochromatization was detected in mature oocytes from directly exposed females. DNA methylation changes showed divergent patterns between male and female tissues, with sperm and testis exhibiting a shift toward hypomethylation, whereas small oocytes, large oocytes, and ovary tissue showed increased hypermethylation.

Epigenetically altered regions were enriched for genes associated with human pathology pathways related to abnormal skeletal morphology, skeletal system abnormalities, brain morphology, and nervous system defects, consistent with phenotypes observed in exposed lineages. Because F3 individuals were not directly exposed to BaP, the methylation signatures detected in this generation are consistent with true transgenerational epimutations transmitted through the germline. Their persistence suggests incomplete erasure during germline reprogramming and/or active maintenance via dysregulated DNA methyltransferases or chromatin remodeling factors.

Collectively, these preliminary results indicate that ancestral BaP exposure induces stable, heritable, and tissue-specific epigenetic reprogramming in O. latipes, providing mechanistic evidence for contaminant-driven transgenerational effects on reproductive epigenetic regulation.

### *Molecular genetic dissection of the mantle in mediating variation in biomineralization and age pathology in clams*

Justin Waletich, Indiana Olson, Matthew Harris and Stephen Treaster
Boston Children's Hospital

The oldest, non-colonial animal, is the clam, with some living more than 500 years. But how do clams with extreme longevity circumvent age-related phenotypes? Our goal is to develop genetic and genomic resources in several clam species to investigate how clams can be leveraged to understand aging and longevity. We focus on mantle epithelia because of its role to initiate and maintain shell biomineralization. To investigate the genetic regulation of differences in aging and longevity, we use comparative genomics in clam species with varying lifespans. Our work in a short-lived clam shows that biomineralization breaks down with age. Large animals show localized changes in mineral aberrations that correlate with deterioration of the mantle epithelial nuclear phenotype. But what mechanisms underly these age-dependent changes? To address this, we have spatially visualized conserved biomineralization markers from a comparative clam mantle snRNA-seq analysis and shown that differential gene sets are associated in the biomineralizing epithelial cells that deteriorate with age. These cellular 'phenotypes' serve as an experimentally measurable readout of tissue aging in the clam mantle to address in experimental approaches targeting longevity genes. We are developing a local species, Spisula solidissima, as a genetic model due to its readily inducible spawning and short development time. Preliminary work shows that electroporation can successfully load gametes with fluorescent molecules for live imaging and dsRNAs to facilitate gene knockdown. Our work lays the foundation for comparative clam genetics and genomics and will help investigate the intersection of shell biomineralization and extreme longevity in these animals.